

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 520 — Past Questions & Materials

by Renance

Course: PEE 520

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 8

Levels: 500

This study pack was compiled and restructured by Renance for PEE 520 students at Federal University of Petroleum Resources, Effurun. It brings together 8 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 8 — PEE 520: NATURAL GAS ENGINEERING TEST 2025/2026

Level 500 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESEARCH

DEPARTMENT OF PETROLEUM ENGINEERING

Pr Son .

2026 aSemester QUIT Mat, No,

1, What is Cooling?

EFFURUN

i

Define refrigeration

What is the Dew point of crude oil?

. Dew point of natural gas is the temperature

Out of the natural

a. TRUE

b. FALSE

and pressure vapour condenses

~ Name two ways the stripping still condenser contribute to the overall process efficiency of the TEG dehydration plant

a.

b.

. In the processing of the reservoir fluid in the separator name the separation technique and mechanism used at the following sections of the separator _

Section Separation Device Separation Mechanism

——— ; |

7. In two one-line sentences give two reasons why the same equipment is called both chiller and evaporator

a.

b.

8. Water vapour turning to ice is

a. Sublimation

b. Evaporation

c. Deposition

a. Condensation

3. Conditioning is a process of

a. Processing crude oil to specific pressure and temperature

b. Processing natural gas to specific pressure and temperature

c. Processing natural gas to have specific liquid content specification

dG. All of the Above

Paper 2 of 8 — PEE 520 NATURAL GAS PROCESSING AND UNIT OPERATIONS 2020

Level 500 • Second Semester • PastQuestion

University of Petroleum Resources, Effurun

Department of Petroleum Engineering

PNG 520 – Natural Gas Processing and Unit Operations

2020 PHND PNG 520 Final Exam

Mat. No. _____ Date: ____9th February,
2020_____

Time: 0000 - 0000 hrs (2 hrs 30 mins)

Basic Instruction

1. This question paper with all charts used in this exam MUST be returned with your answer booklet
2. Answer question no.1 and any other three questions
3. Show all workings on the charts provided on this question paper
4. Ensure this question paper is returned with your answer booklet
5. CONSTANTS:
 - a. 1 lb.mole of natural gas = 379 scf;
 - b. 1 lb.mole of liquid hydrocarbon = 2.76 bbls.
 - c. 1 lb. of water = 20 bbls
 - d. Standard temperature and pressure are 60 OF and 14.7 psia

QUESTION 1 24mark

- a. Define a phase diagram and give its main applications in the oil and gas industry
 - i. Pictorial Temperature-Pressure Diagram illustrating various conditions(T-P) under which different phases of a 2 system(substance) exist
 - a. It enable us know conditions of pressure and temperature under which different phases of substance can exist
 - b. It gives us the tools for the complete understanding of how fluids behave at any of the stages of processing.
 - c. It enable us know the pressure-temperature under which 3 the phases are in equilibrium such that we can employ the equilibrium equation to determine phases quantities and components
 - b. Figure II below is the phase diagram of Stream A from Agbakoba field

which is flowing 390,000 lb.mols/hr of natural gas into the natural gas

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plant clarifier. The reservoir pressure and temperature are 3600 psia and 280 OF respectively

i. What type of reservoir do we have in the Agbakoba field? Explain

It is a gas reservoir

ii. Explain

Because the temperature-pressure condition of the 2 reservoir are the gaseous phase

iii. If the stream of Well A with flow line pressure of 3100 and temperature of 265 is reduced by a JT-Valve having the characteristic chart shown figure III at the gas plant inlet pressure reduction facility before the horizontal clarifier that operates at 600 psia, what will be the operating temperature of the gas plant clarifier? (SHOW YOUR WORK ON THE CHART ALSO)

$$\Delta P = 3100 - 600 = 2500 \text{ psia}$$

From Figure III $\Delta T = 94 \text{ OF. 2}$

$$\Delta \text{Clarifier operating temp.} = 265 - 94 = 171 \text{ OF}$$

Figure III

JT-Valve Characteristic Chart

Δ Temp Drop Vs Press Drop @ Given Press

9

19mark

iv. Sketch and label on Figure II, the production profile(P & T) of the reservoir fluid path from reservoir to the well head and the gas plant clarifier

3600 Reservo

ir

310 Christmas Tree

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Production Profile

from Reservoir to

60 Clarifier 28

17 26

v. Assuming the gas from the flow line is at its liquid dew points, estimate the flow rates out of the gas plant clarifier oil in bbl/day, the water in bbl/day and the gas in mm.scf/day.

1. Calculating for Water

Since 1 lb.mole of natural gas = 379 scf;

$390,000 \text{ lb.mols/hr} = 390,000 \times 379 \times 24 = 3,547,440,000 \text{ scf}$

natural gas/day

Clarifier

inlet P: = 3100; Inlet T: = 265; Water content: = 900 lb. H₂O

/mm.scf

inlet P: = 600; Inlet T: = 171; Water content: = 520 lb. H₂O

/mm.scf

□ Water separated out at the Clarifier = $900 - 520 = 380$

lb.H₂O/mm.scf

Total water in stream = $3,547,440,000 \text{ scf/day} \times 380$

lb.H₂O/mm.scf

1,000,000

= 1,348,027.2 lb. H₂O/day

Now 1 lb. of water = 20 bbls; □ $1,348,027.2 \text{ lb. H}_2\text{O/day} \times 20$

bbl/lb.H₂O

= 26,960,544 bbl/day

For the Clarifier

At 600 psia and 171 OF Oil = 98% and gas = 2%

2. Calculation for Oil

Oil flow rate = $390,000 \text{ moles/hr} \times 98\% \times 2.76 \text{ bbls/lb.mole} \times$

24hr/day

= 25,316,928 bbl/day

3. Calculation for Gas

Gas flow rate = $390,000 \text{ moles/hr} \times 2\% \times 379 \text{ scf/lb.mole} \times$

24hr/day

= $35,474,400,181.92 \text{ scf/day} = 35,474.400$

mm.scf/day

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vi. Sketch and label the gas plant horizontal clarifier in operation if it is water is also expected to be separated from the three-phase gas scrubber.

QUESTION 2

a. Define natural gas processing plant

This is a complex integrated industrial plant designed to purify raw natural gas by separating impurities and various 2 non-methane hydrocarbons fluids out to produce merchantable or pipeline quality dry natural gas (mostly methane).

90

b. Briefly describe and differentiate the processes of purification and 52 conditioning of natural gas

Although the processing of natural gas through the removal of non-hydrocarbon compounds is generally referred to as purification, the word is specifically related to removal of 4 contaminants which is commonly called sweetening. On the other hand, the removal of liquids such as water and hydrocarbon liquids is referred to as conditioning.

c. Briefly describe the operation and main features of the Joule-Thompson Valve

Operation:

□ Stream at high pressure goes through an insulated choke/constriction device and get expanded.

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□ Insulation provides environment for constant enthalpy expansion that gives rise to proportional temperature drop

Features

□ Constant Enthalpy - isenthalpic process

□ No Heat given, No heat taken

4

□ Adiabatic

□ ΔP is proportional to ΔT

d. Sketch and label a block diagram of a typical Low Temperature Separator (LTS) gas plant that employs JT-Valve for natural gas hydrocarbon liquid dewpointing and glycol for water dewpointing.

QUESTION 3

a. Give five(5) main advantages of the tri-ethylene glycol(TEG) in the application of natural gas dehydration

- i. Lower Capital and Operating Cost.
- ii. Decomposition Temperature is Very High (404 0F) 19mark
- iii. □ DEG is 328 0F
- iv. Low Viscosity (Above 70 0F)
- v. Lower Vaporization Loss than EG or DEG
- vi. More Easily Regenerated to Concentration of 98- 99.95% due to its High Boiling and Decomposition Temperature.

b. Sketch and label a detailed diagram of a typical TEG dehydration process plant

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c. In a tabular format, provide all the units of operation in question (b) above with their respective inlet feed, process/Function and products

d. Give detailed description of the processes that take place in the stripping still of the TEG plant and how each process lead to the plant process efficiency.

- i. Condenser: The Rich TEG goes in and out through the tube side to pick heat from the out going steam. In the process, droplets of TEG in stream are stripped off the steam stream.
- ii. The Rich TEG entering the stripping still interact with the upward steam. In the process the upward steam is stripped of its TEG droplets by coalescence. Interaction is provided by the stripping elements
- iii. All these processes lead to process efficiency

19mark

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QUESTION 4

a. Mention five(5) areas of application of heat exchanger in the natural gas industry

i. Potential energy

This is the energy which a body possesses as a consequence of its position in a gravitational field.

ii. Kinetic energy

This is the energy which a body possesses as a consequence of its motion

iii. Internal energy

This is the total energy (potential and kinetic) stored in molecules.

iv. Heat (or thermal) energy

This is kinetic energy due to motion of atoms and molecules. It is energy that is in the process of being transferred from one object to another because of their temperature difference.

v. Radiant energy

This is the energy that propagates through space or through material media in the form of electromagnetic radiation

b. With a simple but well labelled diagram, state the various ways heat is transferred from water in the swimming pool

1

c. What is a Heat Exchanger?

Heat Exchanger is an equipment, a device or contraption that transfer of heat from a hotter medium to a colder medium

d. Explain why the same equipment is referred to as evaporator and chiller.

Because it provide medium for the process fluid to give out heat to become chilled and the for the refrigerant to take in heat to vaporize

e. With an appropriately labelled diagram, give a brief description of the processes that takes place between the natural gas fluid and a refrigerant in an evaporator to produce condensate

i. In the evaporator expanded liquid refrigerant enters in the shell side while the process stream enters into the tube side.

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ii. The refrigerant takes latent heat of vaporization from the process stream and vaporizes. The process fluid giving out

this latent heat loses heat and becomes chilled with liquefiable hydrocarbons being condense out as condensate.

f. In one sentence, differentiate between cooling and refrigeration
Cooling is the reduction of a medium temperature to temperature of the cold condition while refrigeration is the reduction of medium temperature to chilly condition

19mark

QUESTION 5

a. Define phase equilibrium and state four(4) features of a system in equilibrium

Phase equilibrium is a state of coexistence between two phases whereby there is a definite boundary between the two phase 1 while they are maintaining their respective component compositions and properties

State of matter whereby its phases co-exist at a particular and same press and temp.

Features of System in Equilibrium

□ The equilibrium region is the region between bubble point and 3 dew point lines of the phase diagram of a mixture where gas and liquid phases are in equilibrium

□ The equilibrium is maintained by a particular set of pressure and temperature and the quantities and compositions of phases are different for each set of pressure and temperature

□ The equilibrium state exists without change in:

o Pressure

o Temperature

o Molecular attraction

o Molecular repulsion

□ System is at rest

□ No changes taking place

□ Inter-molecular forces of repulsion and attraction are presumably equal

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b. With appropriate and well labelled diagram, briefly describe the process of flash evaporation leading to equilibrium condition.

When liquid is subjected to a sudden pressure change beyond its

bubble pressure, a flash evaporation of the liquid vapour occurs to form a gaseous phase. If the new set of pressure and temperature is maintained, the gaseous and liquid will attain equilibrium condition whereby equilibrium equation can apply to determine their quantities of the phases and their respective components and mole fractions

1

Flash

Evaporati

on

n_L ; x_j

P & T

n ; Z_j Equilibriu

m

c. A gathering line operating at 1050 psi and 190 °F flowing 5360 lb-mole/hour enters into a separator operating at 200 psia and 180 °F. Employing equilibrium equation calculation method with three(3) iterations, determine the approximate flow rate of the separated gas at mm.scf/day and crude oil at bbls/day from the separator. The in-let stream hydrocarbon components and mole fraction composition are given below.

Components Fraction

C3 0.3302

nC4 0.3855

nC5 0.2843

**Attached your working K-value charts for this question to your answer script booklet to obtain maximum points

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3 4 4 4 2

N Componen Mole K- $n_g = 0.006$ $n_g = 0.4$ $n_g = 0.1$ $Y_j = K_j X_j$

o t Fracti Values $X_j =$

Z_j

$X_j =$

Z_j

$X_j =$

Z_j

A on @ $1 + n_g (K_j - 1)$ $1 + n_g (K_j - 1)$ $1 + n_g (K_j - 1)$

G

B 200psia D E F

& 180°F

C

1 C3 0.330 2.00 0.32823 0.23845 0.30018 0.65646

2

2 nC4 0.385 0.80 0.38600 0.41528 0.39337 0.30880

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5

3 nC5 0.284 0.35 0.28541 0.28685 0.30406 0.09990

3

0.99964 1.03907 0.99761 1.06516

 $\sum X_j =$ Assume $n_g = 0.4$ Z_j 0.3302 $C_3; X_j = 0.23586$ $1 + n_g (K_j - 1) = 1 + 0.4 (2 - 1)$ Z_j 0.3855 $C_3; X_j = 0.41902$ $1 + n_g (K_j - 1) = 1 + 0.4 (0.8 - 1)$ Z_j 0.2843 $C_3; X_j = 0.38419$ $1 + n_g (K_j - 1) = 1 + 0.4 (0.35 - 1)$ Assume $n_g = 0.1$ Z_j 0.3302 $C_3; X_j = 0.30018$ $1 + n_g (K_j - 1) = 1 + 0.1 (2 - 1)$ Z_j 0.3855 $C_4; X_j = 0.39337$ $1 + n_g (K_j - 1) = 1 + 0.1 (0.8 - 1)$ Z_j 0.2843 $C_5; X_j = 0.30406$ $1 + n_g (K_j - 1) = 1 + 0.1 (0.35 - 1)$ Assume $n_g = 0.006$ Z_j 0.3302 $C_3; X_j = 0.32823$ $1 + n_g (K_j - 1) = 1 + 0.006 (2 - 1)$ Z_j 0.3855 $C_4; X_j = 0.38600$ $1 + n_g (K_j - 1) = 1 + 0.006 (0.8 - 1)$

$Z_j = 0.2843$

$C_5 = 0.28541$

$1 + n_g (K_j - 1) = 1 + 0.006 (0.35 - 1)$

Since $n_g = 0.006$, $n_L = 1 - 0.006 = 0.994$,

Recall 1 lb-mol of gas = 379 scf

1

$Q_g = 0.006 \times 5360 \times 379 \times 24 = 292,527.36 \text{ scf/day}$

Recall 1 lb-mol of oil = 2.76 bbl

$Q_o = 0.994 \times 5360 \times 2.76 \times 24 = 352,916.123 \text{ bbl/day}$

1

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2.

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QUESTION 6

a. What is unit of operation?

A unit operation is any unit, equipment, device or part carrying out process(es) in the natural gas plant that can be considered to have a single function and producing a desired condition or product(s)

b. Define refrigeration and give five(5) applications of the refrigeration process in the oil and gas industry

a. Refrigeration is the process of temperature reduction to and beyond chilling temperature

Refrigeration Applications

i. Production of LNG

ii. Production of NGL from Natural gas

iii. Nitrogen rejection

iv. Natural gas Water Dewpointing

v. Natural gas HCL Dewpointing

c. With a well labelled diagram, describe the mechanical compression refrigeration system

Refrigerant liquid from storage goes through the expansion valve where the inter-molecular spaces are increased thereby increasing molecular internal energy. It flows into the evaporator where it receives latent heat of vaporization from the process fluid and it vaporizes. The process fluid gets cooled. The refrigerant vapour get compressed at the compressor and it gets hot. The hot compressed vapour get cooled at the condenser and it condenses to liquid and flows into the storage. The process continues again from the storage tank

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d. Sketch and label with detailed description the application of mechanical compression refrigeration system in an LNG production plant

First Refrigeration is Done by Propane to -30°F .

□ Second Refrigeration is Done by Ethylene to -150°F .

□ Third refrigeration is Done by Methane to -250°F

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Paper 3 of 8 — PEE 520 NATURAL GAS PROCESSING AND UNIT OPERATIONS 2018/2019

Level 500 • Second Semester • PastQuestion

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PNG 520 – Natural Gas Processing and Unit Operations

2018/2019 SESSION Final Exam

Mat. No. _____ Date: _____

Time: 0000 - 0000 hrs (3 hrs)

Basic Instruction

1. This question paper with all charts used in this exam MUST be returned with your answer booklet

2. Answer question no.1 and any other three questions

3. Show all workings on the charts provided on this question paper

4. CONSTANTS:

a. 1 lb.mole of natural gas = 379 scf;

b. 1 lb.mole of liquid hydrocarbon = 2.76 bbls.

c. 1 lb.mole of water = 3.512 bbls

d. Standard temperature and pressure are 60 OF and 14.7 psia

QUESTION 1 34

a. Define phase equilibrium and state four(4) features of a system in equilibrium

i. Phase equilibrium is a state of coexistence between two phases whereby there is a definite boundary between the two 1 phase while they are maintaining their respective component compositions and properties

ii. State of matter whereby its phases co-exist at a particular and same press and temp.

Features of System in Equilibrium

□ The equilibrium region is the region between bubble point and 3 dew point lines of the phase diagram of a mixture where gas and liquid phases are in equilibrium

□ The equilibrium is maintained by a particular set of pressure and temperature and the quantities and compositions of phases are different for each set of pressure and temperature

□ The equilibrium state exists without change in:

o Pressure

o Temperature

- o Molecular attraction
- o Molecular repulsion
- ☐ System is at rest
- ☐ No changes taking place
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b. With appropriate and well labelled diagram, briefly describe the process of flash evaporation leading to equilibrium condition

When liquid is subjected to a sudden pressure change beyond its bubble pressure, a flash evaporation of the liquid vapour occurs to form a gaseous phase. If the new set of pressure and temperature is maintained, the gaseous and liquid will attain equilibrium condition whereby equilibrium equation can apply to determine their quantities of the phases and their respective components and mole fractions

1

Flash

Evaporati

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m

c. A gathering line operating at 1050 psi and 190 OF flowing 5360 lb-mole/hour enters into a separator operating at 200 psia and 180 OF. Employing equilibrium equation calculation method with three(3) iterations, determine the approximate flow rate of the separated gas at mm.scf/day and crude oil at bbls/day from the separator. The in-let stream hydrocarbon components and mole fraction composition are given below.

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****Attached your working K-value charts for this question to your answer script booklet to obtain maximum points**

3 4 4 4 2

N Componen Mole K- $n_g = 0.006$ $n_g = 0.4$ $n_g = 0.1$ $Y_j = K_j X_j$
 o t Fracti Values $X_j =$

Z_j

$X_j =$

Z_j

$X_j =$

Z_j

A on @ $1 + n_g (K_j - 1)$ $1 + n_g (K_j - 1)$ $1 + n_g (K_j - 1)$

G

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C

1 C3 0.330 2.00 0.32823 0.23845 0.30018 0.65646

2

2 nC4 0.385 0.80 0.38600 0.41528 0.39337 0.30880

5

3 nC5 0.284 0.35 0.28541 0.28685 0.30406 0.09990

3

0.99964 1.03907 0.99761 1.06516

$\sum X_j =$

Assume $n_g = 0.4$

Z_j 0.3302

C3 ; $X_j = = 0.23586$

$1 + n_g (K_j - 1)$ $1 + 0.4 (2 - 1)$

Z_j 0.3855

C3 ; $X_j = = 0.41902$

$1 + n_g (K_j - 1)$ $1 + 0.4 (0.8 - 1)$

Z_j 0.2843

C3 ; $X_j = = 0.38419$

$1 + n_g (K_j - 1)$ $1 + 0.4 (0.35 - 1)$

Assume $n_g = 0.1$

Z_j 0.3302

C3 ; $X_j = = 0.30018$

$1 + n_g (K_j - 1)$ $1 + 0.1 (2 - 1)$

Zj 0.3855

$C_4; X_j = 0.39337$

$1 + n_g (K_j - 1) 1 + 0.1 (0.8 - 1)$

Zj 0.2843

$C_5; X_j = 0.30406$

$1 + n_g (K_j - 1) 1 + 0.1 (0.35 - 1)$

Assume $n_g = 0.006$

Zj 0.3302

$C_3; X_j = 0.32823$

$1 + n_g (K_j - 1) 1 + 0.006 (2 - 1)$

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Zj 0.3855

$C_4; X_j = 0.38600$

$1 + n_g (K_j - 1) 1 + 0.006 (0.8 - 1)$

Zj 0.2843

$C_5; X_j = 0.28541$

$1 + n_g (K_j - 1) 1 + 0.006 (0.35 - 1)$

Since $n_g = 0.006$, $n_L = 1 - 0.006 = 0.994$,

Recall 1 lb-mol of gas = 379 scf

1

$Q_g = 0.006 \times 5360 \times 379 \times 24 = 292,527.36 \text{ scf/day}$

Recall 1 lb-mol of oil = 2.76 bbl

$Q_{oil} = 0.994 \times 5360 \times 2.76 \times 24 = 352,916.123 \text{ bbl/day}$

1

2.

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QUESTION 2

a. Define NGL and mention the various NGL's that are processed from natural gas in the petroleum Industry

Definition - Natural Gas Liquid

1

- i. NGL's
 - ii. LPG
 - iii. LNG
 - iv. CNG 5
 - v. Condensate
- b. Briefly describe the main difference in PROCESSING between the petroleum products production from the Refinery and GTL plants
- Processing of petroleum products from refinery is mostly and basically physical processes of separations, blending aided with heat with little process of chemical 4 reformation WHILE
- Processing of petroleum products from GTL plants s mostly and basically chemical processes polymerization of methane in large number hydrocarbon compounds of petroleum products
- 1
- c. What is a fractionator? Briefly describe a fractionating process in the production of natural gas of sale quality, LPG and pentane-plus in a typical NGL plant
- i. It is an equipment where distillation process that separates hydrocarbon mixtures into individual products or fractions depending on their volatilities takes place
- 2
- ii. Compounds in a hydrocarbon mixture are separated by heating them to the boiling temperature at which one compound or fractions of the mixture will vaporize.
 - iii. In NGL plant that produces sale natural gas, LPG and pentane plus the plant will be made up a
 1. Deethanizer that removes ethane and methane which constitute the natural gas. The bottom will be Propane 4 plus
 2. Propane plus flows into the Debutanizer that remove that removes Butane and Propane - LPG.
 3. Pentane-plus is removed from the bottom
- d. Sketch and label the above NGL plant with gas arriving from a typical LTS
- plant
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QUESTION 3

a. Define a phase diagram and give its main applications in the oil and gas industry

i. Pictorial Temperature-Pressure Diagram illustrating various conditions(T-P) under which different phases of a 2 system(substance) exist

a. It enable us know conditions of pressure and temperature under which different phases of substance can exist

b. It gives us the tools for the complete understanding of how fluids behave at any of the stages of processing.

c. It enable us know the pressure-temperature under which 3 the phases are in equilibrium such that we can employ the equilibrium equation to determine phases quantities and components

b. Figure II below is the phase diagram of Stream A from Agbakoba field which is flowing 278,000 lb.mols/hr into the gas plant scrubber. The reservoir pressure and temperature are 3000 psia and 275 0F respectively 2

i. What type of reservoir do we have in the Agbakoba field?

It is a wet gas reservoir

Explain

Because the temperature-pressure condition of the reservoir are the gaseous phase while the those of the 2 scrubber are in the two-phase region

ii. If Stream A pressure is reduced by a JT-Valve having the characteristic chart shown figure III to the gas plant scrubber that operates at 1000 psia, what will be the operating temperature of the gas plant scrubber? (SHOW YOUR DEDUCTIONS ON THE CHART ALSO)

$$\Delta P = 3000 - 1000 = 2000 \text{ psia}$$

From Figure III $\Delta T = 70 \text{ 0F. 2}$

$$\Delta \text{ Scrubber operating temp.} = 275 - 70 = 205 \text{ 0F}$$

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iii. Sketch and label on Figure II, the production profile(P & T) of the

reservoir fluid path from reservoir to the gas plant scrubber

R 1

e

sR

ee 2

r s

ve

or

ir v

o

ir

S 2

c S

r c

u r

Figure bIII

u

JT-Valve Characteristic

b Chart

b

□ Temp Drop Vs PresseDrop @ Given b Press

r e

r

P

r

o

d

u

2 c

3 ti

o

n

P

r

b. What are the flow rates of the oil and gas in bbls/day and mm.scf/day respectively

l 1

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i. From Figure II, at 1000psia and 205 0F, we 1% gas and 99% liquid

ii. $Q_g = 0.01 \times 278,000 \times 379 \times 24 = 25,286,880 \text{ scf/day}$ 2

iii. $Q_o = 0.99 \times 278,000 \times 2.76 \times 24 = 18,230,572.8 \text{ bbl/day}$ 2

QUESTION 4

a. Name and define the five(5) main forms of energy 23

i. Potential energy

This is the energy which a body possesses as a consequence of its 1 position in a gravitational field.

ii. Kinetic energy

This is the energy which a body possesses as a consequence of its motion 1

iii. Internal energy

This is the total energy (potential and kinetic) stored in molecules. 1

iv. Heat (or thermal) energy

This is kinetic energy due to motion of atoms and molecules.

It is energy that is in the process of being transferred from 1 one object to another because of their temperature difference.

v. Radiant energy

This is the energy that propagates through space or through material media in the form of electromagnetic radiation 1

b. What is phase transition? In a tabular format, name, define and give example of the basic six phase transitions 1

Pressure-Temperature condition in which molecules of substance changes state of existence or phase

No Phase Phase Example

. Transition Transition

From

1. Melting/ Solid to Liquid Ice melting

Fusion

9

2. Sublimation Solid to Gas Canfor to gas

3 Freezing/ Liquid to Solid Water forming ice

Fusion

4 Evaporation Liquid to Gas Water boiling to

steam

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5 Condensation Gas to Liquid Water vapour to
 rain

6 Deposition Gas to Solid Water vapour to
 snow

c. With appropriate well labelled diagram give a diagrammatic
 explanation of a common everyday situation depicting the various
 phases of transition of matter

3

d. The gas in a cylinder at
 room temperature ($T = 293$
 K), with a volume of 0.096 m^3 is confined by a piston of 500 cm by

3

1500 cm with a weight of 200 N . Atmospheric pressure acts above the
 piston.

i. Determine the pressure of the gas?

2

$P_g A = P_{atm} A + W$. $A = 5 \times 15 = 75 \text{ m}^2$

$\therefore P_g = P_{atm} + \frac{W}{A} = 1.013 \times 10^5 + \frac{200}{75} = 1.013 \times 10^5 + 2.67 =$
 $1.0130267 \times 10^5 \text{ pa}$

A 75

ii. If the gas is heated, expanding it and moving the piston up
 thereby increasing the gas volume three (3) times, how much
 work has the gas done?

Now, recollect that $P_g = 1.0130267 \times 10^5$

And since the gas volume increased 3 times,

$\therefore V = \text{original volume} = 3 \times 0.096 \text{ m}^3 = 0.288 \text{ m}^3$

$\therefore \text{Work done by gas} = W_g = P_g \Delta V = 1.0130267 \times 10^5$
 $\times 0.196$

$= 19,450.113 \text{ J}$

iii. Provide the sketch for i and ii above

2

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Piston

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QUESTION 5

a. Define natural gas processing plant

This is a complex integrated industrial plant designed to purify raw natural gas by separating impurities and various non-methane hydrocarbons fluids out to produce merchantable or pipeline quality dry natural gas (mostly methane).

b. Briefly describe and differentiate the processes of purification and conditioning of natural gas

Although the processing of natural gas through the removal of non-hydrocarbon compounds is generally referred to as purification, the word is specifically related to removal of contaminants which is commonly called sweetening. On the other hand, the removal of liquids such as water and hydrocarbon liquids is referred to as conditioning.

c. Briefly describe the operation and main features of the Joule-Thompson Valve

Operation:

- Stream at high pressure goes through an insulated choke/constriction device and get expanded.
- Insulation provides environment for constant enthalpy expansion that gives rise to proportional temperature drop

Features

- Constant Enthalpy - isenthalpic process
- No Heat given, No heat taken

4

- Adiabatic

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- P is proportional to T

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d. Sketch and label a Deblock diagram

partmen of

t of Petroleum Engineering

ineering Low Temperature

Separator (LTS) plant

PNG 52 that

0 – Natu employs
 ral Gas ProcessingJT-Valve
 and Unit Opfor natural
 erations gas
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 hydrocarbon liquid dewpointing and glycol for water dewpointing. 10
 &RO
 G* DV 6XSSO
 \ * DV
 -7
 9DO
 YH
 / RZ
 Catche 7HP SHUDWXUH
 * DV 6FU
 XEEHU +(; 6W
 DELO
 L]HU
 r 6HSDU DWRU
) HHGVWRFN / 76
 * DV- * DV
 *O\ FRO
 6W
 HDP 3XP S : DU
 P * DV
 / HDQ* O
 \ FRO
 5LFK
 *O
 \ FRO 5LFK *O\ FRO
 2 LO +(;
 +(;
 5HERLOHU *O
 \ FRO 6HSDU DWRU
 5HFRQFHQWUDW
 RU * DV- 5LFK * O
 \ FRO
 / HDQ- 5LFK * O
 \ FRO
 / HDQ* O
 \ FRO 6DOH* DV

1* /

&RQGHQVDW

H

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Paper 4 of 8 — NATURAL GAS ENGINEERING PEE 520 2018

Level 500 • Second Semester • PastQuestion

2018 PNG 520 EXAM QUESTION

One lb-mole conversions:

- Natural gas = 380 scf;
 - Liquid hydrocarbon = 2.85 bbl.
 - Water = 3.1 bbls
2. Standard temperature and pressure are 60 OF and 15 psia.

Question 1

The Genayusuf Petroleum Inc. Central Gas Processing Plant (GCGPL) is being fed with 8790 lb-mole/day of natural gas from the Ugbomro gas field gathering manifold having flow characteristics and parameters as indicated in Table I. The sales natural gas properties of the Genayusuf South-North Transmission Line (GSNTL) are also illustrated in Table I.

Table I

Ex-Well Gas Sale Gas

Component Mole % Mole %

C1. 68.00 87.92

C2 1.40 6.50

C3 0.30 1.51

i-C4 0.45 0.85

n-C4 0.23 0.78

C5+ 2.02 2.02

N2 0.10 0.02

H2S 8.00 0.25

Hydrocarbon liquid 2.00 0.00

H2O(adsorbed) 6.00 0.15

H2O(free liquid) 2.00 0.00

Pressure 2780 psig 985.3 psig

Temperature 189 OF 87 OF

- What will be total daily barrels of free water to be knocked out by the free liquid knock-out at inlet of the gas processing plant?

Mole fraction of Free water = 2% = 0.02 x 8790 lb-mole/day = 175.8 lb-moles/day

Since 1 lb-mole of water = 3.1 bbls, total free water = 3.1 x 175.8 = 544.98 bbls/day

- Define unit of operation

Any unit, equipment, device or part carrying out process(es) in the natural gas plant that can be considered to have a single function and producing a desired condition or product(s)

- With an appropriate choice of processing equipment/devices/units of operation, give a sketch of a well-labelled process flow block diagram for the GCGPL

Page 1 of 14

d. In a tabular format, list all the equipment/devices/units of operation employed in the gas processing with their respective thermodynamic flow parameters, inlet feedstock and outlet products.

Equipment Press in Press Temp Temp. Feed stock Product out

psig out in out OF

psig OF

Free Liquid Knock- 2780 2775 187 185 Well Stream • Wet Gas

out Natural Gas • Water and HCL Free Liquids

Classifier/Separator 2775 2770 185 183 Water and HCL • Wet Gas

Free Liquids • Water Free Liquids

• HCL Free Liquids

Gas-Gas HEX 2770 2770 183 150 Wet Gas • Lower Temp Wet Gas

1000 1000 36 87 Chilled NG • Sale Temp NG

Expansion 2770 1000 150 36 Lower Temp • Chilled Sour Wet NG

Refrigeration Unit

LT Separator 1000 990 36 35 Chilled Wet NG • Chilled Sour Water Wet NG

• Chilled NGL

Sweetening Unit 1000 990 87 87 Chilled Sour Water • H₂S

Wet NG • Chilled Sweet Water Wet

NG

Dehydration Unit 990 985.3 87 87 Chilled Sweet • Chilled Sweet Dry NG with

Water Wet NG Nitrogen

Nitrogen Rejection 985.3 985.3 87 87 Chilled Sweet Dry • Nitrogen

unit NG with Nitrogen • Chilled Sweet Dry Sale NG

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Question 2

a. Briefly explain the phase being “homogeneous and physical distinct part of a system with definite bounding surfaces (interfaces)”

A phase being homogenous means that it is the same in nature in all parts of that phase and it is Physically different(Distinct Part) of the substance (System) with a Definite Physical Bounding Surfaces (Interface) different from any other phase of the same substance.

b. What is the difference between phase behaviour and phase diagram?

Phase behaviour is the changes in phase exhibited by the natural gas when subjected to different temperature-pressure conditions while the phase diagram is the pressure temperature diagrammatic representation of the various temperature- pressure conditions at which the natural gas exist in different phases

c. State and describe three(3) main applications of phase behaviour in the natural gas processing

Phase behaviour is the exhibition of different phases of natural gas when subjected to changes in temperature and pressure is applied in natural gas processing in the followings

a. NGL Extraction

The knowledge of the phase behaviour of the NG enables one to tell the temperature and pressure under which the NGL will condense out of the NG and to design the process and

b. LNG Production

The Knowledge of when NG will turn to liquid is used to design the process of its liquefaction and production of LNG

c. Hydrate Prevention and control

The knowledge of when water will condense out NG is used to set operating temperature of natural gas transmission line above the NG water dew point such that water condensation will be prevented and hydrate formation will be prevented

d. Refrigeration Dehydration

The knowledge of when water will condense out NG is used to design the operating temperature of the refrigeration process meant for NG dehydration

e. Prevention of Slugging/Two-Phase Flow in NG Transmission Lines

The knowledge of when water will condense out NG is used to set operating temperature of transmission lines to prevent water condensation and thus preventing slugging/two-phase flow

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d. Define phase equilibrium and state four(4) features of a system in equilibrium

Phase equilibrium is a state of coexistence between two phases whereby there is a definite boundary between the two phase while they are maintaining their respective component compositions and properties

Features of System in Equilibrium

- The equilibrium state exists without change in:
 - o Pressure
 - o Temperature
 - o Molecular attraction
 - o Molecular repulsion
- System is at rest
- No changes taking place
- Inter-molecular forces of repulsion and attraction are presumably equal

e. Briefly describe the process of flash evaporation leading to equilibrium condition

When liquid is subjected to a sudden pressure change beyond its bubble pressure, a flash evaporation of the substance vapour occurs to form a gaseous phase. If the new set of pressure and temperature is maintained, the gaseous and liquid will attain equilibrium condition whereby equilibrium equation can apply to determine their quantities of the phases and their respective components and mole fractions

Flash

Evaporation n_L ; x_j

P & T

n ; Z_j Equilibrium

Question 3

a. Define equilibrium calculation and give its two(2) uses in natural gas processing

i. This is calculation used to determining the fractions, quantities and components of liquid and vapour that are in equilibrium after flash evaporation

ii. Used for determination of flow rate of gas and liquid from the separator

iii. Used to determine optimum separator

iv. operating pressure and temperature condition

b. A gathering line operating at 750 psia and 112 OF flowing at 1570 lb-mole/hour enters into a separator operating at 200 psia and 180 OF. Employing equilibrium equation calculation method with three(3) iterations:

i. Determine the component mole-fractions of the separated gas and liquid,

The in-let stream component and mole fraction composition are given below.

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Components Fraction

C3 0.4132

i-C4 0.3825

i-C5 0.2043

No Component Mole K-Values $\square$

$\square = 0.7$

$\square \square = 0.85$

Fraction @ =

Z_j

Z_j

$Y_j = K_j X_j$

$(K - 1)$

X_j

$(K - 1)$

X_j

$1 + n_{g,j} \quad 1 + n_{g,j}$

200 psia

and F G

D E

A B 100°F.

C

1 Propane 0.4132 2.00 0.2431 0.2234 0.4468

2 i-Butane 0.3825 1.05 0.3696 0.3669 0.3853
 3 i-Pentane 0.2043 0.41 0.3480 0.4098 0.1673
 0.9607 1.0001 0.9994

$\sum X_j =$

Iteration #1; α

$\alpha = 0.7$

$\alpha = 0.4132$

$\alpha_3; \alpha_3 = 1 + \alpha(\alpha_3 - 1) = 1 + 0.7(2 - 1)$
 $= 0.2431$

α_3

$\alpha_3 = 0.3825$

$\alpha_4; \alpha_4 = 1 + \alpha(\alpha_4 - 1) = 1 + 0.7(1.05 - 1)$
 $= 0.3696$

α_4

$\alpha_4 = 0.2043$

$\alpha_5; \alpha_5 = 1 + \alpha(\alpha_5 - 1) = 1 + 0.7(0.41 - 1)$
 $= 0.3480$

α_5

Iteration #1; α

$\alpha = 0.85$

$\alpha = 0.4132$

$\alpha_3; \alpha_3 = = = 0.2234$
 $1 + \alpha$

$-(\alpha$

$\alpha_3 - 1) 1 + 0.85(2 - 1)$

$\alpha = 0.3825$

$\alpha_4; \alpha_4 = 1 + \alpha(\alpha_4 - 1) = 1 + 0.85(1.05 - 1)$
 $= 0.3669$

α_4

$\alpha_4 = 0.2043$

$\alpha_5; \alpha_5 = = = 0.4098$
 $1 + \alpha$

$-(\alpha$

$\alpha_5 - 1) 1 + 0.85(0.41 - 1)$

ii. Determine the approximate flow rate of the separated gas in scf/day from the separator

$\alpha_3 \alpha_4 \alpha_5$

$\alpha = 0.85$, gas flow rate = $0.85 \times 1570 \text{ lb-mole/hour} \times 24 = 32,028 \text{ lb-mole /day}$

But 1 lb-mole of gas = 380 scf(given) $\therefore 32,028 \text{ lb-mole /day} = 32,028 \text{ lb-mole /day} \times 380$
 scf/lb-mole
 $= 12,170,640 \text{ scf/day}$

iii.. Determine the approximate flow rate of the separated liquid hydrocarbon in bbl/day from the separator

□□□□□ □

□ = 0.85; □h□□ □

□ = 1.0 – 0.85 = 0.15 and as 1 lb-mole of HCL = 2.85 bbl,

□ Separated HCL = 0.15 x 1570 x 24 x 2.85 = 16,108.2 bbl/day

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Question 4

a. Define mass transfer and give two examples of processes of mass transfer in oil and gas processing

Definition: This is the transfer of mass into or out of a substance or transfer of a chemical compound from one phase to another

Examples:

- a. Evaporation of liquid to gas
- b. Diffusion of solute from high concentration to low concentration
- c. Absorption of one or more compounds from gas phase to a liquid phase – TEG dehydration

b. With a well labelled diagram(s), give a detailed description of mass transfer absorption process of ammonia from polluted air by water

Gas absorption (also called gas absorption, gas scrubbing or gas washing) essentially involved the transfer of materials from the gas phase to the liquid phase.

Three basic components are involved in gas absorption;

The medium containing the solute - needs purification

The solute - impurity or desirable component - need to be removed

The absorbent(solvent) - the solute removing agent

For Example;

Mixture of air and ammonia that needs to be purified for pollution requirement by removal of ammonia with water

Three components are

Air - medium needing purification

Ammonia - Solute to be removed

Water - absorbent(solvent) to remove ammonia

NH₄ H₂O

Ammonia

polluted air

Purified Air

Absorption Process

Water

Rich Water

Page 6 of 14

- c. Sketch and label a TEG gas absorption plant with its basic components
- d. With a well-labelled diagram give a brief description of Nitrogen Rejection Unit by Solvent Absorption Process
 - ◆ In the Solvent Absorption process, Methane and heavier hydrocarbons are separated from the nitrogen by a lean solvent.
 - ◆ The methane and heavier hydrocarbons are flashed off from the rich solvent by depressurizing stream.
 - ◆ The lean solvent from the flash regeneration process is returned back to the top of the methane absorber.
 - ◆ In the process, Nitrogen is normally vented into the atmosphere with small percentage of methane or re-injected for EOR projects.

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Question 5

- a. Name and define the five(5) main forms of energy

1. Potential energy

This is the energy which a body possesses as a consequence of its position in a gravitational field (e.g., water behind a dam).

2. Kinetic energy

This is the energy which a body possesses as a consequence of its motion (e.g., wind blowing across or a wind generator). It is dependent upon an object's mass and velocity (e.g., moving water versus moving air).

3. Internal energy

This is the total energy (potential and kinetic) stored in molecules.

4. Heat (or thermal) energy

This is kinetic energy due to motion of atoms and molecules. It is energy that is in the process of being transferred from one object to another because of their temperature difference.

5. Radiant energy

This is the energy that propagates through space or through material media in the form of electromagnetic radiation.

- b. With appropriate well labelled diagram give a diagrammatic explanation of the various phases of transition of matter

- c. In a tabular format, name, define and give example of each phase transition

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No. From Required To

State Energy/Process State

1. Solid Melting Liquid

/Fusion

2. Solid Sublimation Gas

3 Liqui Freezing/Fusion Solid

d

4 Liqui Evaporation Gas

d

5 Gas Condensation Liquid

6 Gas Deposition Solid

d. Calculate the heat capacity of 1050 grams of water if 5030 J of heat is required to raise its temperature from 10 °C to 75 °C.

$$Q = m \times C_p \Delta T = 5030$$

$$C_p = \frac{Q}{m \Delta T} = \frac{5030}{1050(75 - 10)} = 0.0737 \text{ J/g } ^\circ\text{C}$$

$$m \Delta T = 1050(75 - 10)$$

e. Calculate the rate of heat transfer across 1.5mm thick metal with thermal conductivity constant of 240 J/m s K when heat was used to raise its temperature from 60 °C to 75 °C.

$$K = 240 \text{ J/m s K}; L = 1.5\text{mm} = 0.0015\text{m}; \text{ and } \Delta T = 75 - 60 = 15$$

Now

$$\Delta T = 75 - 60 = 15$$

$$Q' = \frac{KA \Delta T}{L} = \frac{240 \times A \times 15}{0.0015} = 2.4 \times 10^6 \text{ J/(s m}^2\text{)}$$

$$L = 0.0015$$

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Question 6

a. Employing the principle of thermal energy transfer, define and explain the process of heat transfer

Basic Principle of Thermal Energy Transfer

I. Heat transfer is the flow of thermal energy driven by thermal non-equilibrium (i.e. a non-uniform temperature field, existence of temperature gradient).

II. Thermal energy of a body or matter is related to the temperature of the matter and the higher the body temperature, the greater the matter thermal energy.

III. When two bodies are at different temperatures, thermal energy transfers from the one with higher temperature to the one with lower temperature - Heat always transfers from hot to cold.

Definition of Heat transfer

Heat transfer is the process of exchange of thermal energy through a body or between bodies when there is a temperature difference - a thermal energy in transit due to temperature difference.

Explanation of Heat Transfer Process

The basic requirement for heat transfer is the presence of temperature difference which is the driving force for heat transfer, just as the voltage difference is the driving force for electric

current flow and pressure difference is the driving force for fluid flow.

b. With the help of a single diagram give the definitions of the four(4) main modes of heat transfer

Various heat Transfer Modes – The swimming pool example

c. With appropriate example, give five(5) applications of heat transfer in oil and gas industry

i. Boiling - Heat transferred to rich glycol to boil off absorbed water

ii. Refrigerant Refrigeration - Process fluid gives to refrigerant to vaporize and gets itself chilled HCL condenses out HC Dewpointing

iii. Cold process fluid absorbed heat from inlet feedstock to obtained sale value temperature

iv. Boiled steam from TEG reboiler used to heat rich TEG to reduce its required heat need for boiling at the reboiler thereby increasing plant process efficiency

v. Heat is transferred to water saturated absorber beds in order to regenerate solid dessicant.

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vi. Thermal processing is applied to force a temperature change in the system that enables or disables some material transformation such as absorbed water from glycol, absorbed contaminants(H₂S, CO₂, Mercury, etc.) from amine solutions, adsorbed water and contaminants from solid desiccants, breaking down of water and oil emulsion, etc.

vii. Thermal control is applied to regulate within fixed established bounds, or to control in time within a certain margin, the temperature of a system to secure process or plant proper and correct functioning.

viii. Fluids out of process equipment requiring specific temperature status are guaranteed through heat transfer equipment

d. Define heat exchanger and state its three(3) main benefits in natural gas processing

An equipment where heat is exchanged between two or more fluids

◆ Process Efficiency

◆ Energy Conservation

◆ Reduces Maintenance

e. Sketch and label a typical shell and tube heat exchanger heating cold gas from turbo-expander with steam generated from a rich TEG reboiler.

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Paper 5 of 8 — Natural Gas Engineering PEE 520 2017

Level 500 • Second Semester • PastQuestion

Federal University of Petroleum Resources, Effurun

Department of Petroleum Engineering

Final Exam

Course: Natural Gas Engineering II

Date:

Time: 0000 - 0000 hrs (3 hrs)

Name: _____ ID # _____

Surname First Name Initial

Basic Instruction

1. Answer question no. 1 and any other four questions
2. Show all workings on the charts provided where applicable

285

160

132 194

1

6

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1. The Prema gas processing central node is being fed with natural gas from the gas well having flow characteristic flow parameters as indicated in Table I. Table III illustrates properties of the sales natural gas.

a. What will be total daily pounds of free liquids to be knocked by the free liquid knock-out at inlet of the gas processing plant?

Free water = $180 \times 10 = 1,800 \text{ lb/day}$

Free HC liquid = $180 \times 6.8 = 1,224 \text{ lb/day}$

Total Daily Free liquid removal = $3,024 \text{ lb/day}$

b. If a JT-valve with the characteristic chart given in figure I below were to be employed to reduce the pressure of the natural gas free of liquids to its sale condition,

i. What will be the JT-Valve outlet temperature of the gas?

Inlet Pressure = $190 \times 14.504 = 2,755.76 \text{ psi}$

Outlet Pressure = $76.8 \times 14.504 = 1,113.91 \text{ psi}$

Pressure drop across JT-Valve = $1,641.85 \text{ psi}$

From JT-Valve Chart Temperature Drop = 62 OF

Outlet Temperature = $90 \text{ OC}; T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 9/5 + 32 = 90 \times 9/5 + 32 = 194 \text{ OF}$

Outlet Temperature = $194 - 62 = 132 \text{ OF}$

ii. How many pounds of water will be removed by the JT-Valve?

From Wehe Chart

Gas Water content at 194 OF and 2,755.76 psi = 285 lb/mmscf

Gas Water content at 132 OF and 1,113.91 psi = 160 lb/mmscf

Water removed = (285 - 160) lb/mmscf x 180 mm.scf/day =
22,500lb/day

c. Using the JT-valve in (a) above and with your choice of appropriate processes, equipment and devices, give a block process flow diagram required to process and treat this gas to its sale quality and condition

2

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62

Molecular

Sieve

(H₂O Removal)

HEX JT- Sep Molecular

Inlet FLK

Valv arat Sieve

Gas O (H₂S + CO₂)

or

Sale

Separat Gas Condens

or ate H₂S +

Storage CO₂

(H₂O +

Disposal

2. Water

a. Briefly explain what is meant by a phase being homogeneous, Treatment physically distinct with definite bounding interfaces

Homogeneous: - All and every part or portion of the phase is the same in content, composition and physical appearance

Physically distinct with definite bounding interfaces: - All phases of the same substance are distinct and different in appearance, physical structure and proper ties

b. Using basic principle of thermodynamics and intermolecular theory, briefly explain the process of molecules of water (being pure substance) moving from liquid state to vapour state upon pressure reduction

Thermodynamically, expansion gives rise to a higher state of energy.

In this higher state of energy, molecules gain energy and vibrate at higher rate and this

gives rise to greater physical intermolecular distance(greater space).

With more energy with further expansion, the moles are at higher energy level to move in the vapour state.

3

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To move to vapour state, molecules will take latent heat of vaporization from the environment.

This latent energy is supplied by the energy giving source BUT because it is taken internally, it DOES NOT reflect in the expansion increase(pressure reduction) in the measuring device. Hence until all water turns to vapour, no pressure reduction will be experienced.

c. Stream A in figure II below is flowing at 12,000 lb.mols/day from a reservoir with 3750 psia and 300 OF

i. What is the phase of the stream in the reservoir?

-----Gas-----

ii. Assuming a lb.mol of gas is 350 scf(irrespective of pressure and temperature) and lb.mol of liquid is 0.025 bbl, determine the flow rate of gas and oil flowing from a separator operating at 100 psia and 100

0

F

From the figure at 100 psia and

100 OF, fluid is 94% gas

□ Gas flow rate =

$0.94 \times 12,000 \times 350$

$= 3,948,000$

scf/day

And Liquid flow rate =

$0.06 \times 12,000 \times 0.025 = 18$

bbl/day

3.

a. Mention five(5) areas of application of heat exchanger in the natural gas industry

i. Oil and gas processes involving energy utilization, thermal processing, and thermal control.

ii. Thermal processing is applied to force a temperature change in the system that enables or disables some material transformation such as absorbed water from glycol, absorbed contaminants(H₂S, CO₂, Mercury, etc.) from amine solutions, adsorbed water and contaminants from solid desiccants, breaking down of water and oil emulsion, etc.

- iii. Thermal control is applied to regulate within fixed established bounds, or to control in time within a certain margin, the temperature of a system to secure process or plant proper and correct functioning.
- iv. Fluids out of process equipment requiring specific temperature status are guaranteed through heat transfer equipment
- v. Fluids Going into Process Equipment Requiring Specific Temperature Status
- vi. Use of Waste Energy for Efficient Processing and Economic Purposes

4

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vii. Fluids Out of Well Requiring different Temperature Processing Conditions

viii. Storage Conditions Requiring Different Temperature From Processing Conditions.

ix. Heat is used to boiled off absorbed water from glycol and solid desiccants in natural gas dehydration process

x. Heat is used to break off H₂S and CO₂ molecules in natural gas sweetening

xi. Heat is used to break emulsion in crude oil emulsion treatment

xii. Natural Gas Liquefaction is achieved by transferring heat out of natural gas through the process of Refrigeration

b. With a simple but well labelled diagram, state the various ways heat is transferred from water in the swimming pool

c. What is a Heat Exchanger?

Heat Exchanger is an equipment, a device or contraption that transfer of heat from a hotter medium to a colder medium

d. Explain why the same equipment is referred to as evaporator and chiller.

Because it provide medium for the process fluid to give out heat to become chilled and the for the refrigerant to take in heat to vaporize

e. With an appropriately labelled diagram, give a brief description of the processes that takes place between the natural gas fluid and a refrigerant in an evaporator to produce condensate

i. In the evaporator expanded liquid refrigerant enters in the shell side while the process stream enters into the tube side.

ii. The refrigerant takes latent heat of vaporization from the process stream and

vaporizes. The process fluid giving out this latent heat loses heat and becomes chilled with liquefiable hydrocarbons being condensed out as condensate.

f. In one sentence, differentiate between cooling and refrigeration

Cooling is the reduction of a medium temperature to temperature of the cold condition while refrigeration is the reduction of medium temperature to chilly condition

4.

a. Define LNG and CNG; Give the similarity and differences in terms of content and processing

LNG is Liquefied Natural Gas and CNG is Compressed Natural Gas

5

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Similarities

- i. Both are liquid natural gases
- ii. Processing is usually turning gas to liquid

Differences are

i.

While LNG refers to basically methane mixture with very minute propane and butane, CNG be mostly propane, butane, mixture of propane and butane or mixture of these and methane

ii.

LNG is normally through the process of cryogenic refrigeration while CNG is normally through the process of compression

b. In a bullet format, give a brief description of the LNG production process with the Propane Pre-Cool Mixed Refrigerant natural gas liquefaction process

- Incoming gas is sent through pressure equalization process where the inlet gas is reduced to the plant pressure. Condensate from this is sent for storage
- The gas is then treated for contaminants such as CO₂, H₂S, Mercury, etc. at the sweetening sub-plants
- The gas is then sent for dehydration for water removal and Water is sent for water treatment plant and disposal
- The gas is sent for propane pre-cool refrigeration where all liquid hydrocarbons are removed and sent for condensate storage and the fractionator.
- Methane and ethane are fed into the Main Cryogenic Mixed Refrigerant Heat Exchanger

□ Inside this, the mixed-Refrigerant takes heat from the methane to turn it to Liquid.

□ A specified proportion of the propane and butane are fed in the gas inside the heat Exchanger from the fractionator to increase the LNG heating value.

□ At the fractionator, propane, butane and NGL are produced.

c. Given the sales gas with the component composition in question 1 above, provide a detailed well labelled process flow diagram of the propane pre-cool mixed refrigerant cryogenic refrigeration process to produce LNG from the gas.

Considering specification requirement for Hydrogen Sulphide 1 grains/Ccf3; Carbon Dioxide 2% by volume; mercaptan 2 grains/Ccf3; Nitrogen 2% by volume and Oxygen 0.4% by volume. It means the gas meets specifications for LNG for the above contaminants and does not require treatment

LNG to

Storage

Propane

Pre- Main

Inlet Pressure Cooled Mixed-

Gas Equalizati Compressi Refrigeran

on on t

refrigerati

6

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Separat

or Ethane

(H₂O + Fractionatio Propan

n Butane

e

Pentan

Condens e+

ate

Surge

d. In a tabular format, provide all the units of operation in question (c) above with their respective inlet feed, process/function and product

No Unit of Operation Feed Function Product

1 Pressure equalization Raw gas Provide plant Stream with constant

Facility pressure pressure

2 Separator Free liquid Separation Water and Condensate

3 Condensate Surge Condensate Storing Stored condensate
 4 C3-precooled Plant pressure Cool to remove HC-
 Refrigeration Gas HC-liquid liquid
 Methane

5 Mixed refrigerant HEX Methane, C3 Methane Sale Spec. LNG
 &C4 Liquefaction

6 Fractionator HC-Liquid Fractionation C3, C4, C2, C5+

5.

a. Give five(5) main advantages of the tri-ethylene glycol(TEG) in the application
 of natural gas dehydration

i. Lower Capital and Operating Cost.

ii. Decomposition Temperature is Very High (404 0F)

iii. □ DEG is 328 0F

iv. Low Viscosity (Above 70 0F)

v. Lower Vaporization Loss than EG or DEG

vi. More Easily Regenerated to Concentration of 98- 99.95% due to
 its High Boiling and Decomposition Temperature.

b. Sketch and label a detailed diagram of a typical TEG dehydration process
 plant

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c. In a tabular format, provide all the units of operation in question (b) above
 with their respective inlet feed, process/Function and products

No Unit of Operation Feed Function Product

1 Contactor Raw gas and Medium for gas Dry gas, Rich TEG
 Lean TEG dehydration

2 Stripping Still Rich TEG, Separation, Steam and TEG Droplets
 Steam with Condensation and
 TEG droplets cooling

3 Reboiler Rich TEG + Rich TEG Steam, Lean TEG
 Heat Stripping and
 Boiling, Water

Vaporization

4 Flash Drum Rich TEG Rich TEG Gas + Rich TEG
 Expansion and
 Gas Flashing

5 Heat Exchanger Rich TEG + Heat Exchange Cooler Lean TEG and
 Lean TEG Hotter Rich TEG

6 Pump LP Lean TEG Increase Pump HP Lean TEG
 Press

Carbon Filter Lean TEG Filtering -Removal Lean TEG w/o Grease
of HC Solid

Filter Lean TEG Filtering -Removal Lean TEG w/o Solids
of Solid

Surge Tank Lean TEG With- Stored Lean TEG for
Holding(Storage) pumping

d. Give detailed description of the processes that take place in the stripping still
of the TEG plant and how each process lead to the plant process efficiency.

i. Condenser: The Rich TEG goes in and out through the tube side to pick
heat from the out going steam. In the process, droplets of TEG in
stream are stripped off the steam stream.

ii. The Rich TEG entering the stripping still interact with the upward
steam. In the process the upward steam is stripped of its TEG droplets
by coalescence. Interaction is provided by the stripping elements

iii. All these processes lead to process efficiency

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Paper 6 of 8 — PEE 520 NATURAL GAS ENGINEERING 2018I2019

Level 500 • Second Semester • PastQuestion

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PNG 521 – Advance Drilling Engineering

2017/2018 Session Final Examination

Mat. No. _____ Date: _____

Basic Instructions

1. Duration: 3 hours
2. Answer Question 1 and Four(4) Others
3. Some relevant charts and graph sheets are provided

QUESTIONS 1

The figure below is the pore pressure and fracture pressure gradient plot of formations where the Sand Taminutu was discovered with attractive reserves at 11500 ft

- a. On the figure, give a preliminary casing design to drill to the 2 reservoir, sketching on the figure the casings and their respective depths.

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- b. Calculate for the production casing, the Burst design pressure and Collapse design pressure using API Design Factors.

- i. Calculate Burst Production casing

Pore pressure at 11,500ft = 10.5 ppg;

$$\square P_{burst} = 0.052 \times 10.5 \times 11,500 = 6,272$$

psi

- ii. Calculate Collapse pressure

Drilling mud weight at 11,500 ft = 10.75 ppg.

$$\square P_{collapse} = 0.052 \times 10.75 \times 11,500 = 6,428.5$$

psi

- c. Taking into consideration ONLY both the Burst and Collapse pressures, design and select the long thread coupling (LTC) 7 inches production casing giving its grade and nominal weight.

- i. With $P_{burst} = 6,272$ psi; collapse pressure of 6,428.5

and LTC coupling, from the table

HCL-80 casing is the only that satisfies all design conditions.

Casing Grade - HCL -80

Casing size - 7"

Casing nominal weight - 26 lb/ft

Burst rating - 7,200 psi

Collapse rating - 7,800 psi

d. What will be mud pressure with which you will drill at 6,500 ft with an under-balance of 0.55 ppg?

Mud weight at 6,500 ft = 10.3 ppg. With an under-balance of 0.55 ppg, drilling mud weight will 10.3 - 0.55 = 9.75 ppg.

$$P_{\text{mud}} = 0.052 \times 9.75 \times 6,500 = 3,295.5 \text{ psi}$$

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QUESTIONS 2

a. With basic definitions ONLY, differentiate between drilling methods and drilling technologies 1

i. Drilling Methods consist of the body of techniques and equipment employed to break rock into fragments or core in drilling (constructing or making hole) in the subsurface rock in the oil and gas industry while

ii. Drilling Technology involves the application of -2 equipment, tools and drilling methods in the construction of a well.

b. What is abrasion drilling method?

i. Drilling method in which the rock is worn, gorged, peeled and chipped off by rotating action upon the rock of a shoe or bit armed with some harder material buttons or teeth.

c. How does the roller-cone bit drill in rotary drilling technology?

i. Upon rotation of the bit while drilling fluid is being circulated and weight being put on the bit, the bit teeth on the cones cut the formation into pieces called cuttings. These cuttings are simultaneously swept and

carried from the wellbore bottom up the annulus to the surface.

d. Briefly describe the percussion drilling technology and give five (5) advantages of drilling with it.

i. This involves the application of other equipment, tools and percussion drilling method to pound the rock into fragments small enough to be readily removed from the hole by the circulating drilling fluid.

ii. ADVANTAGES

i. Improve hole geometry,

ii. Increased ROP and fastest drilling in hard formation

iii. Reduces drillstring stresses,

iv. Generates better and larger cuttings which yields a better representation for geological study

v. Lower drilling cost per foot substantially

vi. Most efficient drilling method in hard formations – generally 2-3 times higher than conventional drilling method and 7.3 times higher in medium-hard granite

vii. Because it involves less contact time with rock - only 1 or 2 percent of total operational drilling

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time - bit has less abrasion and hence a longer bit life.

viii. Less use of other bottom hole assembly – saving equipment rental cost

ix. Less hole deviation and easier control of deviation problem for straight hole drilling.

e. Define the rotary percussion drilling technology and describe how it drills

i. This is drilling technology that simultaneously implores both abrasion(rotary) and percussion drilling methods to drill the formation.

ii. The rotary percussion uses a mud motor mechanism to offer rotation while a percussion mechanism-commonly called the hammer - is located directly behind the drill

- bit reciprocating piston-driven "hammer" to energetically drive a heavy drill bit into the rock.
- iii. The bit faces remains in constant contact with the wellbore bottom while the piston behind the bit face impacts the face repeatedly fracturing the formation.
 - iv. Drilling fluid(mud, water or air) at up to 180 bar delivery pressure is used to power the impact mechanism of the hammer at a high frequency rate.
 - v. As the fluid then leaves the hammer, it loses the pressure and keeps a low flush velocity which is still adequate to bring any cuttings to the surface and clean the bore
 - vi. This percussive action increases the instantaneous weight on bit (WOB), even though the mean WOB may be kept at or below typical levels for drilling
 - vii. This action results in increased penetration rate of the drill bit, longer drill bit life due to less WOB application, and unplanned borehole deviation may be decreased.

QUESTIONS 3

- a. Define drilling service contract

The Drilling Service Contract is an agreement between the 1 operator and the service company to drill and complete a well OR to render certain specialized services within the drilling and completion processes.

- b. State and define four main types of contracts used in the service industry of the oil and gas industry

- i. Drilling Contracts

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This is the contract agreement between the operator and the service company to drill and complete a well.

The nature, major parts and operation are dealt with in subsequent sections.

- ii. Seismic Acquisition Agreements

These are agreements/licences/contracts between Seismic companies and government or land owner

to carry out geophysical and seismic surveys.

iii. Well services contracts

These are contracts between service companies and operator to carry out specific service(s) in or during the drilling of the well for the operator

iv. Master Service Contract

These are “Master Service Agreements” formulated by “big” oil and gas producing companies or service companies when they expect an ongoing relationship with a reliable service company Or the oil company

c. List and give a brief explanation of the five(5) main components of drilling service contract

i. The works cope/specifications

This clause specifies the

n Wellbore diameter of each section of the well

□ Depth of each section of the well

□ The drilling muds to be used in each section

□ The deviational and directional requirement of each section

□ The equipment, technology and services to be employed in each section.

ii. Payment/Fees Terms

A clause that specifies the rate (cost) for the service and payment methods. It can be per day, per foot, Footage rate, Day work rate, Turnkey rate, Combination and Alliance.

Other fees include Standby rates, Mobilization rates, Commissioning

Rates and Decommissioning rates

iii. Liability or Risk Allocation Clauses

A clause that allocates liability or risk to each of the partners in the drilling contract. Insurance is normally

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required to cover the cost of these liability and risk allocations.

iv. Waiver or Consequential Damages clauses

A clause that absolves a party from the consequential damages arising from actions of the other(s) that may affect him or the other(s).

v. Force Majeure Terms

A clause that takes care of an unexpected event that crucially affects somebody's ability to do something and can be used in law as an excuse for not having carried out the terms of an agreement (formal)

d. In a tabular format define footage rate and day work rate payment terms of the drilling contract, give their respective advantages, disadvantages and areas of application

Payment Method	Definition	Advantage	Disadvantage	Areas of Application
Footage Rate	is per Operator Desire for Where very rate foot and pays exactly achieving little or payment for service higher nothing is is on total rendered. footage known number of Increases sometimes about the feet drilled Rate of leads to less area drilling payment to Service safety company. measures.	Put operator It also on his toe decreases and brings quality the best out of him		
Day Rate	is per Easy Payment Where a lot Work day in determinati could be is known in work and on of work made for the area pay is to be paid days that do made on for. not yield the total Increases any footage. number of operator's Makes days supervision service to ensure company work is done lazy and for the day play pranks			

n

Footage Rate is per Operator Desire for Where very rate foot and pays exactly achieving little or payment for service higher nothing is

is on total rendered. footage known

number of Increases sometimes about the

feet drilled Rate of leads to less area

drilling payment to

Service safety

company. measures.

Put operator It also

on his toe decreases

and brings quality

the best out

of him

Day Rate is per Easy Payment Where a lot

Work day in determinati could be is known in

work and on of work made for the area

pay is to be paid days that do

made on for. not yield

the total Increases any footage.

number of operator's Makes

days supervision service

to ensure company

work is done lazy and

for the day play pranks

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QUESTIONS 4

a. In a sequential format, list the processes involved in well design from Data Acquisition and review to Pre-spudding Meeting. 19

- i. Data Acquisition and Review
- ii. Data Analysis
- iii. Well Design Meeting
- iv. Well Design
- v. Circulate the Design for Comment
- vi. Write the Drilling Program
- vii. Circulate the Program for Comment
- viii. Pre-spud Meeting

b. Give brief description of what wells' hole section summaries in drilling and explain how they are used in well design.

i. Hole Section Summaries are summaries created for each section drilled in previous wells and they contain large quantities on the

i. Interval of formation – (2000 – 2900 ft)

ii. Formation

1. Lithology
2. Compressive strength
3. Competence – loose, caving, hard, etc
4. Stickiness, tight hole, drag, etc on pull
5. Problem associated with drilling through

iii. Drilling parameters

1. WOB
2. RPM
3. GPM
4. Torque
5. Tripping condition

iv. BHA

1. Packed (Locked)
2. Build angle
3. Drop angle
4. Mud motor

v. Drilling fluid

1. WBM

2. KCL

3. OBM

4. Water, etc

vi. Comments on

1. Casing/cement jobs,

2. Over gauge hole from logs, and

3. Any problems logging or other worthwhile comments

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ii. Uses in Well Design

i. Comparisons can be quickly made in the same formations between several wells.

ii. The knowledge of the lithologies and problem areas, with notes on compressive strengths (derived from sonic logs) and other relevant information help in designing with

1. Preventive measures.

2. Better bit selection

3. BHA with high Drillability in the section

4. Better selection of RPM and GPM

5. Better mud properties

c. Design (to scale) the horizontal section of the well with surface location at S15000 and E75000 and the target location at N55000 and E15000. What is the direction (in quadrant notation) of the well and its horizontal displacement?

$$\square E/W = 75,000 - 15,000 = 60,000$$

$$\square N/S = 55,000 + 15,000 = 70,000$$

$$HD = \sqrt{N^2 + E^2} = \sqrt{(70000)^2 + (60000)^2} = 92,195.45 \text{ ft}$$

$$A = \tan^{-1} \frac{60,000}{70,000} = \tan^{-1} 0.857142 = 40.6 = N40.6W$$

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QUESTIONS 5

b. Give reasons/justifications for the liability or risk allocation clauses in drilling contracts

□ Drilling is a high risky process associated with huge cost.

□ Problem arising from drilling are associated with substantial damage to property and in personal injuries.

□ Problem may also occur at thousands of feet down hole where direct observation is difficult, the nature and cause may be difficult to ascertained and appropriated.

□ This clause endeavour to eliminate uncertainty with regard to responsibility for consequences relating to problems that arise by ensuring that responsibilities for damages are clearly stated.

□ The goal is to ensure that when various scenarios come to pass the partners responsible for damages are well enunciated, and to make sure that there are neither gaps in insurance coverage, nor double coverage.

□ This also avoids protracted litigation

c. Briefly state the purpose of the indemnification clause

The purpose is to exempt (indemnify) parties from penalties 19 and liabilities.

d. Give detailed explanation of what “Knock for Knock” stands for, how it works, the condition under which it does not apply and sate reason for its non-application

Both operator and contractor separately assumes all risk associated with its own equipment and personal, regardless of

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fault, and agrees to indemnify the other party for claims arising out of injuries suffered to their respective employees and properties.

No waiver or indemnity clause will be allowed by law for damages arising from acts of “gross negligence” or breach of safety rules and regulations from any party

e. State and briefly explain three(3) contractor’s indemnities and three(3) operator’s indemnities

Contractors’ Indemnities

□ The infringement of patents or other industrial

property rights arising out of or in connection with the Work or Company's use of the Work, except where such infringement necessarily arises from the use of Company Provided Items or Company's instructions

□ The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is caused by the negligence or breach of duty (whether statutory or otherwise) of Contractor Group.

□ The pollution occurring on the premises of Contractor Group or originating from the property and equipment of Contractor Group located above the rotary table, arising from or relating to the performance of the Work.

Operators' Indemnities

□ The loss of or damage to any well or hole

□ The blowout, fire, explosion, cratering or any uncontrolled well condition (including the cost to control a wild well and the removal of debris)

□ The damage to any reservoir, geological formation or underground strata or the loss of oil or gas therefrom

□ The use of radioactive tools in relation to the Work or any contamination resulting therefrom (including retrieval and/or containment and clean up) - This applies regardless of any form of liability, whether strict or by negligence, in whatever form, on the part of Contractor group.

□ The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is not caused by the negligence or breach of duty (whether statutory or otherwise) of Company.

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QUESTIONS 6

A vertical well with N72456.4 and E21995.6 was drilled to top of reservoir sand Bemba at 12500 ft when it blew out. To kill the well, a relief well was sited at N61986.8 and E18794.7 The relief well was

planned to kick-off at 750 ft and build angle to 45° and thereafter maintain inclination to intersect the vertical well at the top of Bemba. We only have a 30 bent sub available at site 19

a. What is the total vertical depth of the relief well?

i. Horizontal Displacement

$$HD = \sqrt{N^2 + E^2} = \sqrt{(72456.4 - 61986.8)^2 + (21995.6 - 18794.7)^2}$$

$$= \sqrt{(10,469.6)^2 + (3200.9)^2} = \sqrt{109,612,524.16 + 10,245,760.81}$$

$$= \sqrt{119,858,284.97} = 10,947.981 \text{ ft}$$

ii. Radius of Curvature $R = 180 \times CL = 180 \times 100 = 1909.9 \text{ ft}$

iii. $\pi \times BUR = 3.142 \times 3$

iv. Build TVD = $R \times \sin I = 1909.9 \times \sin 45 = 1350.5 \text{ ft}$

v. Build Section HD = $R \times (1 - \cos I) = 1909.9(1 - \cos 45) = 559.4 \text{ ft}$

vi. Tangent Section TVD = $THD - BHD = 10947.981 - 559.4 = 10,388.581 \text{ ft}$

vii. $\tan I = \tan 45$

viii. Total TVD = $750 + 1,350.5 + 10,388.581 = 12,489.081 \text{ ft}$

b. What is the total measured depth of the relief well?

i. Length of Build Section (BUAL)

$$= 45 \times 100 = 45 \times 3.142 \times 1909.9 = 1,500 \text{ ft}$$

ii. Length of Tangent Section

$$= \text{Tangent Section TVD} = 10,388.581 = 14,691.672 \text{ ft}$$

$\cos I = \cos 45$

iii. Total Measured Depth = $750 + 1500 + 14,691.672 = 16,941.672 \text{ ft}$

c. Sketch the vertical section of the relief well.

Casing Properties Table

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Paper 7 of 8 — PEE 520 2017 Exam PQ

Level 500 • Second Semester • PastQuestion

Federal University of Petroleum Resources, Effurun

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Final Exam

Course: Natural Gas Engineering II

Date:

Time: 0000 - 0000 hrs (3 hrs)

Name: _____ ID # _____

Surname First Name Initial

Basic Instruction

1. Answer question no. 1 and any other four questions
2. Show all workings on the charts provided where applicable

285

160

132 194

1

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1. The Prema gas processing central node is being fed with natural gas from the gas well having flow characteristic flow parameters as indicated in Table I. Table III illustrates properties of the sales natural gas.

a. What will be total daily pounds of free liquids to be knocked by the free liquid knock-out at inlet of the gas processing plant?

Free water = $180 \times 10 = 1,800$ lb/day

Free HC liquid = $180 \times 6.8 = 1,224$ lb/day

Total Daily Free liquid removal = 3,024 lb/day

b. If a JT-valve with the characteristic chart given in figure I below were to be employed to reduce the pressure of the natural gas free of liquids to its sale condition,

i. What will be the JT-Valve outlet temperature of the gas?

Inlet Pressure = $190 \times 14.504 = 2,755.76$ psi

Outlet Pressure = $76.8 \times 14.504 = 1,113.91$ psi

Pressure drop across JT-Valve = 1,641.85 psi

From JT-Valve Chart Temperature Drop = 62 OF

Outlet Temperature = 90 OC; $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 9/5 + 32 = 90 \times 9/5 + 32 = 194$ OF

Outlet Temperature = $194 - 62 = 132$ OF

ii. How many pounds of water will be removed by the JT-Valve?

From Wehe Chart

Gas Water content at 194 OF and 2,755.76 psi = 285 lb/mm scf

Gas Water content at 132 OF and 1,113.91 psi = 160 lb/mm scf

Water removed = (285 - 160) lb/mm scf x 180 mm.scf/day =
22,500lb/day

c. Using the JT-valve in (a) above and with your choice of appropriate processes, equipment and devices, give a block process flow diagram required to process and treat this gas to its sale quality and condition

2

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Molecular

Sieve

(H₂O Removal)

HEX JT- Sep Molecular

Inlet FLK

Valv arat Sieve

Gas O (H₂S + CO₂)

or

Sale

Separat Gas Condens

or ate H₂S +

Storage CO₂

(H₂O +

Disposal

2. Water

a. Briefly explain what is meant by a phase being homogeneous, Treatment physically distinct with definite bounding interfaces

Homogeneous: - All and every part or portion of the phase is the same in content, composition and physical appearance

Physically distinct with definite bounding interfaces: - All phases of the same substance are distinct and different in appearance, physical structure and proper ties

b. Using basic principle of thermodynamics and intermolecular theory, briefly explain the process of molecules of water (being pure substance) moving from liquid state to vapour state upon pressure reduction

Thermodynamically, expansion gives rise to a higher state of energy.

In this higher state of energy, molecules gain energy and vibrate at higher rate and this

gives rise to greater physical intermolecular distance(greater space).

With more energy with further expansion, the moles are at higher energy level to move in the vapour state.

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To move to vapour state, molecules will take latent heat of vaporization from the environment.

This latent energy is supplied by the energy giving source BUT because it is taken internally, it DOES NOT reflect in the expansion increase(pressure reduction) in the measuring device. Hence until all water turns to vapour, no pressure reduction will be experienced.

c. Stream A in figure II below is flowing at 12,000 lb.mols/day from a reservoir with 3750 psia and 300 OF

i. What is the phase of the stream in the reservoir?

-----Gas-----

ii. Assuming a lb.mol of gas is 350 scf(irrespective of pressure and temperature) and lb.mol of liquid is 0.025 bbl, determine the flow rate of gas and oil flowing from a separator operating at 100 psia and 100

0

F

From the figure at 100 psia and

100 OF, fluid is 94% gas

□ Gas flow rate =

$0.94 \times 12,000 \times 350$

= 3,948,000

scf/day

And Liquid flow rate =

$0.06 \times 12,000 \times 0.025 = 18$

bbl/day

3.

a. Mention five(5) areas of application of heat exchanger in the natural gas industry

i. Oil and gas processes involving energy utilization, thermal processing, and thermal control.

ii. Thermal processing is applied to force a temperature change in the system that enables or disables some material transformation such as absorbed water from glycol, absorbed contaminants(H₂S, CO₂, Mercury, etc.) from amine solutions, adsorbed water and contaminants from solid desiccants, breaking down of water and oil emulsion, etc.

- iii. Thermal control is applied to regulate within fixed established bounds, or to control in time within a certain margin, the temperature of a system to secure process or plant proper and correct functioning.
- iv. Fluids out of process equipment requiring specific temperature status are guaranteed through heat transfer equipment
- v. Fluids Going into Process Equipment Requiring Specific Temperature Status
- vi. Use of Waste Energy for Efficient Processing and Economic Purposes

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vii. Fluids Out of Well Requiring different Temperature Processing Conditions

viii. Storage Conditions Requiring Different Temperature From Processing Conditions.

ix. Heat is used to boiled off absorbed water from glycol and solid desiccants in natural gas dehydration process

x. Heat is used to break off H₂S and CO₂ molecules in natural gas sweetening

xi. Heat is used to break emulsion in crude oil emulsion treatment

xii. Natural Gas Liquefaction is achieved by transferring heat out of natural gas through the process of Refrigeration

b. With a simple but well labelled diagram, state the various ways heat is transferred from water in the swimming pool

c. What is a Heat Exchanger?

Heat Exchanger is an equipment, a device or contraption that transfer of heat from a hotter medium to a colder medium

d. Explain why the same equipment is referred to as evaporator and chiller.

Because it provide medium for the process fluid to give out heat to become chilled and the for the refrigerant to take in heat to vaporize

e. With an appropriately labelled diagram, give a brief description of the processes that takes place between the natural gas fluid and a refrigerant in an evaporator to produce condensate

i. In the evaporator expanded liquid refrigerant enters in the shell side while the process stream enters into the tube side.

ii. The refrigerant takes latent heat of vaporization from the process stream and

vaporizes. The process fluid giving out this latent heat loses heat and becomes chilled with liquefiable hydrocarbons being condense out as condensate.

f. In one sentence, differentiate between cooling and refrigeration

Cooling is the reduction of a medium temperature to temperature of the cold condition while refrigeration is the reduction of medium temperature to chilly condition

4.

a. Define LNG and CNG; Give the similarity and differences in terms of content and processing

LNG is Liquefied Natural Gas and CNG is Compressed Natural Gas

5

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Similarities

- i. Both are liquid natural gases
- ii. Processing is usually turning gas to liquid

Differences are

i.

While LNG refers to basically methane mixture with very minute propane and butane, CNG be mostly propane, butane, mixture of propane and butane or mixture of these and methane

ii.

LNG is normally through the process of cryogenic refrigeration while CNG is normally through the process of compression

b. In a bullet format, give a brief description of the LNG production process with the Propane Pre-Cool Mixed Refrigerant natural gas liquefaction process

- Incoming gas is sent through pressure equalization process where the inlet gas is reduced to the plant pressure. Condensate from this is sent for storage
- The gas is then treated for contaminants such as CO₂, H₂S, Mercury, etc. at the sweetening sub-plants
- The gas is then sent for dehydration for water removal and Water is sent for water treatment plant and disposal
- The gas is sent for propane pre-cool refrigeration where all liquid hydrocarbons are removed and sent for condensate storage and the fractionator.
- Methane and ethane are fed into the Main Cryogenic Mixed Refrigerant Heat Exchanger

□ Inside this, the mixed-Refrigerant takes heat from the methane to turn it to Liquid.

□ A specified proportion of the propane and butane are fed in the gas inside the heat Exchanger from the fractionator to increase the LNG heating value.

□ At the fractionator, propane, butane and NGL are produced.

c. Given the sales gas with the component composition in question 1 above, provide a detailed well labelled process flow diagram of the propane pre-cool mixed refrigerant cryogenic refrigeration process to produce LNG from the gas.

Considering specification requirement for Hydrogen Sulphide 1 grains/Ccf3; Carbon Dioxide 2% by volume; mercaptan 2 grains/Ccf3; Nitrogen 2% by volume and Oxygen 0.4% by volume. It means the gas meets specifications for LNG for the above contaminants and does not require treatment

LNG to

Storage

Propane

Pre- Main

Inlet Pressure Cooled Mixed-

Gas Equalizati Compressi Refrigeran

on on t

refrigerati

6

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Separat

or Ethane

(H₂O + Fractionatio Propan

n Butane

e

Pentan

Condens e+

ate

Surge

d. In a tabular format, provide all the units of operation in question (c) above with their respective inlet feed, process/function and product

No Unit of Operation Feed Function Product

1 Pressure equalization Raw gas Provide plant Stream with constant

Facility pressure pressure

2 Separator Free liquid Separation Water and Condensate

3 Condensate Surge Condensate Storing Stored condensate
 4 C3-precooled Plant pressure Cool to remove HC-
 Refrigeration Gas HC-liquid liquid
 Methane

5 Mixed refrigerant HEX Methane, C3 Methane Sale Spec. LNG
 &C4 Liquefaction

6 Fractionator HC-Liquid Fractionation C3, C4, C2, C5+

5.

a. Give five(5) main advantages of the tri-ethylene glycol(TEG) in the application
 of natural gas dehydration

i. Lower Capital and Operating Cost.

ii. Decomposition Temperature is Very High (404 0F)

iii. □ DEG is 328 0F

iv. Low Viscosity (Above 70 0F)

v. Lower Vaporization Loss than EG or DEG

vi. More Easily Regenerated to Concentration of 98- 99.95% due to
 its High Boiling and Decomposition Temperature.

b. Sketch and label a detailed diagram of a typical TEG dehydration process
 plant

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c. In a tabular format, provide all the units of operation in question (b) above
 with their respective inlet feed, process/Function and products

No Unit of Operation Feed Function Product

1 Contactor Raw gas and Medium for gas Dry gas, Rich TEG
 Lean TEG dehydration

2 Stripping Still Rich TEG, Separation, Steam and TEG Droplets
 Steam with Condensation and
 TEG droplets cooling

3 Reboiler Rich TEG + Rich TEG Steam, Lean TEG
 Heat Stripping and
 Boiling, Water

Vaporization

4 Flash Drum Rich TEG Rich TEG Gas + Rich TEG
 Expansion and
 Gas Flashing

5 Heat Exchanger Rich TEG + Heat Exchange Cooler Lean TEG and
 Lean TEG Hotter Rich TEG

6 Pump LP Lean TEG Increase Pump HP Lean TEG
 Press

Carbon Filter Lean TEG Filtering -Removal Lean TEG w/o Grease
of HC Solid

Filter Lean TEG Filtering -Removal Lean TEG w/o Solids
of Solid

Surge Tank Lean TEG With- Stored Lean TEG for
Holding(Storage) pumping

d. Give detailed description of the processes that take place in the stripping still
of the TEG plant and how each process lead to the plant process efficiency.

i. Condenser: The Rich TEG goes in and out through the tube side to pick
heat from the out going steam. In the process, droplets of TEG in
stream are stripped off the steam stream.

ii. The Rich TEG entering the stripping still interact with the upward
steam. In the process the upward steam is stripped of its TEG droplets
by coalescence. Interaction is provided by the stripping elements

iii. All these processes lead to process efficiency

8

Paper 8 of 8 — PEE520 NATURAL GAS PROCESSING AND UNIT OPERATIONS

Level 500 • Second Semester • Handout

Federal University of Petroleum Resources

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Course:

PNG 520

Natural Gas Processing and Unit

Operations

Ogbon Natural Gas Processing & Unit Slide #1

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Lecturer:

Ogbarode N. Ogbon

Ogbon Natural Gas Processing & Unit Slide #2

Lesson IV - III

Natural Gas Processing and Unit Operations

(Absorption Processes)

Ogbon Natural Gas Processing & Unit Slide #3

Absorption Process Unit Operations

□ Natural Gas Absorption Dehydration

□ Definition

□ Absorption is a process whereby a targeted compound is removed from a process stream by a liquid agent.

□ Absorption is used for natural gas dehydration, sweetening and conditioning by way of NGL removal

□ Liquid Desiccant

□ Liquid that absorbs.

No Process Type of Liquid Component

. Desiccant Removed

1 NGL Lean Oil NGL

Extraction

2 Sweetening Amines, Physical Solvents, H₂S, CO₂, etc. etc.

□ Ethylene Glycol - EG

3 Dehydration □ Di-ethylene Glycol □ DEG

Water or Water

□ Tri-ethylene Glycol □ TEG Vapour

□ Tetra-ethylene Glycol □ T4EG

Ogbon Natural Gas Processing & Unit Slide #4

Absorption Process Unit Operations

□ NGL Recovery Oil Absorption Process

□ This involves the absorption of NGL from the natural gas stream in an absorber column by lean oil of about 150 MW at about 100 °F

□ The rich oil gets its propane and lighter components removed at a depropanizer column to recombine with the methane from the absorber

□ NGL is removed at the fractionating column as an overhead product while lean oil is recovered at the bottom and recycled

□ The unit is capable of recovering 75% butane and 85 – 90% C5+

Ogbon Natural Gas Processing & Unit Slide #5

Absorption Process Unit Operations

□ NGL Recovery Oil Absorption Process

Process Flow Diagram for NGL Lean Oil Absorption Recovery Process

Ogbon Natural Gas Processing & Unit Slide #6

Adsorption Process Unit Operations

□ Natural Gas Absorption Sweetening Processes

□ Acid gases such as H₂S and CO₂ are removed by absorption sweetening process

□ Liquid desiccants commonly used in acid gas removal include

□ An aqueous amine

□ Mixed solutions

□ Physical solvents

□ Other Patent Formulations

Ogbon Natural Gas Processing & Unit Slide #7

Adsorption Process Unit Operations

□ Natural Gas Absorption Sweetening Processes

□ Mixed Solutions

□ Mixture of Amine, Physical Solvent and Water.

□ Capable of High Acid Gas Loading.

□ Also Absorb Organic Sulfur

□ Examples of Mixed Solutions

- Sulfinol
- Optisol
- Flexsorb
- Ucarsol

Ogbon Natural Gas Processing & Unit Slide #8

Adsorption Process Unit Operations

- Natural Gas Absorption Sweetening Processes
- Physical Solvents
- Organic Liquids That Absorb H₂S and CO₂ at High Pressure and Ambient or Low Temperature
- Used for Bulk Removal of CO₂
- Capable of Being Regenerated Without Heat.
- Capable of Drying Gas Simultaneously During Regeneration.
- Examples of Physical Solvents
- Fluor Solvent
- Selexol
- Purisol
- Rectisol

Ogbon Natural Gas Processing & Unit Slide #9

Absorption Process Unit Operations

- Natural Gas Absorption Sweetening Processes
- Aqueous Amine Solution
- Solutions are Regenerated.
- Used to Remove Large Amount of Sulfur and CO₂
- Types of Amine Solutions Used
- Monoethanolamine □ MEA
- Diethanolamine □ DEA
- Triethanolamine □ TEA
- Diglycolamine □ DGA
- Methyldiethanolamine □ MDEA
- Other Patent Formulations.
- Ucarsol
- Flexsorb
- Amine Guard, etc

Ogbon Natural Gas Processing & Unit Slide #10

Absorption Process Unit Operations

- Natural Gas Amine and Mixed Solutions Sweetening Processes
- .
- Characteristics

- Process is Continuous.
- Operating Cost Per Pound of Sulfur Removed is Low.
- Amine Solution Composition Can Be Tailored to the Sour Gas Composition for More Efficient Removal.
- Generally, Amine Bases are too Weak for Organic Sulfur Contaminants Such as RSH, COS and RSSR(Disulfide Aliphatic) but Large Amount can be Removed With Addition of Physical Solvent.
- Major Disadvantages :
 - High Capital Investment Compared to Batch Process.
 - High Operating and Maintenance Cost.
 - Some Processes Require License or Royalty Fees.

Ogbon Natural Gas Processing & Unit Slide #11

Absorption Process Unit Operations

Sweetening Liquids Molecular Structures

General Amines

.....

Hindered Amines

.....

.....

.....

Physical Solvents

Ogbon Natural Gas Processing & Unit Slide #12

Absorption Process Unit Operations

Comparison of Gas Sweetening Solutions

Ogbon Natural Gas Processing & Unit Slide #13

Absorption Process Unit Operations

- Natural Gas Absorption Sweetening by Aqueous Amine
- The sour gas flows from the separator into the lower part of the about 20 - 24 trays absorber or contactor while the lean amine solution is pumped into the absorber near the top.
- As the solution flows down from tray to tray, it gets into intimate contact with the sour gas that flows upward through the liquid on each tray and H₂S, CO₂ and total sulfur content of the natural gas are removed.
- The rich solution flows out of the contactor through the flash drum, where most dissolved hydrocarbon gas and some acid gas get flashed off. It then goes through the heat exchanger where it picks up heat from the hot regenerated lean solution stream before going into the regeneration

section.

□ The liberated acid gas at the regeneration section and any hydrocarbon gas not flashed off in the flash drum leave the still at the top, together with some solvent and a lot of water vapor.

Ogbon Natural Gas Processing & Unit Slide #14

Absorption Process Unit Operations

□ Natural Gas Absorption Sweetening

For monoethanolamine (MEA) denoted as RNH₂ the reaction is



Ogbon Natural Gas Processing & Unit Slide #15

Absorption Process Unit Operations

□ Natural Gas Absorption Sweetening by Aqueous Amine

□ Acid Gases Are Burned or Sent for Processing.

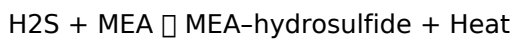
□ Hot Lean Amine Solution From Reboiler Heats Cold Foul Amine Solution From Contactor.

□ It Is Cooled at the Cooler to Ambient Temperature Before Entering Surge Tank.

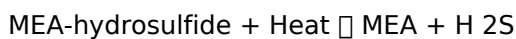
□ From the Surge Tank It Is Pumped Into the Contactor

Basic Reactions

Contaminant Removal



Regeneration



Ogbon Natural Gas Processing & Unit Slide #16

Absorption Process Unit Operations

Basic Components of the Amine Sweetening Plant

□ Inlet Scrubber Removes:

□ Entrained Liquids:

□ Commonly Slugs of Produced (Saline) H₂O.

□ Contaminate Amine Solution.

□ Causes Corrosion.

□ Causes Solvent Losses.

□ Causes Amine Foaming.

□ Causes Equipment Fouling.

□ Corrosive Inhibitors.

□ Well Treating Chemicals.

- Be Located as Close as Possible to the Absorber.
- The Absorber Tower
- Either Trays or Packings
- May Contain Water Wash in the Upper Section to Remove Amine From Outlet Gas.

Ogbon Natural Gas Processing & Unit Slide #17

Absorption Process Unit Operations

Basic Components of the Amine Sweetening Plant

- Regeneration System
- Flash Tank
- Flashes Out HC Absorbed From the Gas in the Absorber on Pressure Reduction.
- Provides Enough Time for Degassing of the Rich Amine Solution.
- Lean/Rich Amine Heat Exchanger.
- Preheats Rich Glycol Before Still and Reboiler.
- Cools the Lean Amine Before Absorber.

□ Stripping Still

- Either Tray or Packing Type
- Reboiler □ Direct Fire Heater
- Reflux Condenser □ Air Cooled.
- Amine Cooler □ Air Cooled

Ogbon Natural Gas Processing & Unit Slide #18

Absorption Process Unit Operations

□ Amine Sweetening Plant

Sweet

Gas Out

Outlet Scrubber

Sour

Gas in

Inlet Scrubber

Ogbon Natural Gas Processing & Unit Slide #19

Absorption Process Unit Operations

- Dehydration by Glycol Absorption
- Advantages of Tri-ethylene Glycol (TEG)
- Lower Capital and Operating Cost.
- Decomposition Temperature is Very High (404 0F)
- □ DEG is 328 0F
- Low Viscosity (Above 70 0F)

- Lower Vaporization Loss than EG or DEG
- More Easily Regenerated to Concentration of 98-99.95% due to its High Boiling and Decomposition Temperature.

Ogbon Natural Gas Processing & Unit Slide #20

Absorption Process Unit Operations

- TEG Dehydration Plant
- Basic Components
- Inlet Scrubber □ Removes Entrained or Free Water, Oils or Dissolved Hydrocarbons and salt water
- Contactor □ Creates room for maximum glycol/gas interaction for water absorption
- Glycol pump □ pumps lean glycol up contactor
- Glycol cooler □ cools contact hot inlet lean glycol
- Stripping still □ Strips reboiler inlet rich glycol and outlet stream
- Stripping Still Condenser □ Condenses glycol out of exit steam
- Reboiler □ Reconcentrates the rich glycol by boiling off the water
- Heat Exchanger □ Heats up rich glycol with lean glycol while cooling it.
- Filter and Strainer □ Removes solids and grease

Ogbon Natural Gas Processing & Unit Slide #21

Absorption Process Unit Operations

TEG Dehydration Plant Process Flow Diagram

Ogbon Natural Gas Processing & Unit Slide #22

Absorption Process Unit Operations

Ogbon Natural Gas Processing & Unit Slide #23

Absorption Process Unit Operations

□ Contactor Absorber

□ .

Jimkevnic@gmail.com

Ogbon Natural Gas Processing & Unit Slide #24

Absorption Process Unit Operations

□ Typical Commercial Trays

□ .

□ .

Standard Flexitray Valv

Flexitray Valve Tray

Sieve Tray Bubble-Cap Tray

Ogbon Natural Gas Processing & Unit Slide #25

Absorption Process Unit Operations

- Packed Columns

- .

- Intalox Packing □

Packed Elements

Ogbon Natural Gas Processing & Unit Slide #26

Absorption Process Unit Operations

- Re-Boiler and Stripping Still

- Operating @ 118 °C and 100

mbarg

- To achieve required purity of glycol

- To minimize glycol decomposition

al Direct Fired Reboiler Temp. Profile

Ogbon Natural Gas Processing & Unit Slide #27

Absorption Process Unit Operations

- Tri-Ethylene Glycol(TEG)

Dehydration

- Factors for Consideration in TEG Plant Operations.

- Water, Hydrocarbon Liquids and Lubricating Oils in Gas Require That an Efficient Separator Be Installed Upstream of Absorption Tower.

- Water With High Concentration of Minerals in Gas May Crystallize Over a Long Period and Fill the Reboiler With Solid Salts.

- Highly Concentrated Glycol May Be Difficult to Pump at Low Temperature Because of Their High Viscosity.

Ogbon Natural Gas Processing & Unit Slide #28

Absorption Process Unit Operations

- Tri-Ethylene Glycol(TEG)

Dehydration

- Factors for Consideration in TEG Plant Operations.

- In Cold Weather Regions, Glycol Lines Have Tendency to Solidify If Not in Use, Therefore Continuous Circulation is Required.

□ Sudden Surges Should Be Avoided in Starting and Shutting Down the Plant to Avoid Occurrence of Large Carry-over Losses of Glycol.

□ Foreign Matter Such As Dirt, Iron Oxide, etc. Can Contaminate Glycol.

□ Decomposition of Glycol May Occur If Overheated.

□ The Presence of Oxygen and Hydrogen Sulphide May Cause Corrosion.

Ogbon Natural Gas Processing & Unit Slide #29

Associated Gas Gathering Unit Operations

□ Major Processes.

□ Compression □ Sales

Gas

□ Cooling

□ Condensate Extraction

□ Inlet □ Dehydration

□ Gas □ Dehydration

n

□ Compression □ Compression □ Compression

n n □

□ (12.3 bar) (70 bar)

□ (30 bar)

□ Condensa □ Condensa

□ Condensa

□ Cooling □ Cooling te

□ Cooling te

te

Extraction Extraction

Extraction

Ogbon Natural Gas Processing & Unit Slide #30

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Course:

PNG 520

Natural Gas Processing and Unit

Operations

Ogbon Natural Gas Processing & Unit Slide #1

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Ogbarode N. Ogbon

Ogbon Natural Gas Processing & Unit Slide #2

Lesson IV - I

Natural Gas Processing and Unit Operations

(Refrigeration Processes)

Ogbon Natural Gas Processing & Unit Slide #3

Sub-Topics

□ Natural Gas Conditioning and Unit Operations

□ Refrigeration Processes

□ NGL Unit Operation

□ LPG Unit Operation

□ Fractionation Process

□ LNG Process

Ogbon Natural Gas Processing & Unit Slide #4

Overview of Natural Gas Conditioning and Unit

□ Natural Gas Conditioning Operations

□ The natural gas processing plant is a complex integrated industrial plant designed to purify raw natural gas by separating impurities and various non-methane hydrocarbons fluids out to produce merchantable or pipeline quality dry natural gas (mostly methane).

□ The separation or removal of the impurities (contaminants) is commonly called sweetening and sometimes referred to as purification.

□ Conditioning on the other hand involves the removal of hydrocarbon liquids and water.

□ Liquefiable hydrocarbons - C₃+ - in addition to major portion of ethane content of the natural gas are removed to the barest minimum remaining majorly methane to constitute sales natural gas.

□ Water that reduces the heating value of the gas, in addition to causing corrosion and forming hydrates, is also removed in conditioning.

□ The various components removed and the processes used to remove them are shown in the table in the next slide.

Ogbon Natural Gas Processing & Unit Slide #5

Overview of Natural Gas Conditioning and Unit Operations

□ The Natural Gas Plant Processes and Products

□ Nitrogen

Ogbon Natural Gas Processing & Unit Slide #6

Overview of Natural Gas Conditioning and Unit Operations

□ Natural Gas Plant Unit Operations

□ For each of the processes, there is a unit of operation(device, equipment, or semi-plant) made up of its equipment.

□ A typical natural gas processing plant unit operations process diagram is laid out in the figure in next slide.

□ Note that the number of units of operations that constitute the entire processing plant is determined by the gas constituent components and the level of treatment required to meet customer and regulatory requirements.

□ A unit operation therefore, is any part of processes in the natural gas plant that can be considered to have a single function and producing a desired condition or product(s). Common unit operations in the natural gas plant include the followings;

□ Separation Processes

□ Sweetening Processes

□ Dehydration Processes

□ Heat Exchanger Processes

□ NGL Extraction

□ Fractionation Processes

□ Nitrogen Rejection Unit(NRU)

□ Figure below indicates all processes the natural gas can undergo depending on its contents

Ogbon Natural Gas Processing & Unit Slide #7

Overview of Natural Gas Conditioning and Unit

□ Natural Gas Plant Unit Operations

Operations

Ogbon Natural Gas Processing & Unit Slide #8

Sub-Topic

Refrigeration Processes

Ogbon Natural Gas Processing & Unit Slide #9

Refrigeration Process Unit Operations

□ Refrigeration Process

□ Refrigeration is the process of temperature reduction.

□ Depending on the level of temperature reduction, its application will include:

□ NGL Extraction

- LPG Extraction
- LNG Liquefaction
- Nitrogen Removal
- The methods of refrigeration employed in the natural gas processing include the followings;
- Mechanical Compression Refrigeration
- J-T Valve Refrigeration
- Turbo-Expander Refrigeration
- Cascade Refrigeration

Ogbon Natural Gas Processing & Unit Slide #10

Refrigeration Process Unit Operations

- NGL Mechanical Refrigeration Recovery Unit

Ogbon Natural Gas Processing & Unit Slide #11

Turbine Expansion Refrigeration

System

Sales Gas

Inlet Gas

Separator

Gas-Gas

Heat Exchanger

Expansion Liquid to

Turbine Stabilization

Ogbon Natural Gas Processing & Unit Slide #12

- Propane-Ethylene- Methane Cascade

Refrigeration System

- First Refrigeration is Done by □ Second Refrigeration is Done

Propane to -30 0F. by Ethylene to -150 0F.

- Third refrigeration is Done by

Methane to -250* 0F

□ .

□ .

Chiller Chiller Chiller

-30 0F -150 0F -250 0F

Water Propane

Condenser Evaporator Ethylene

Evaporator

Ogbon Natural Gas Processing & Unit Slide #13

Refrigeration Process Unit Operations

□ Pressure Expansion Refrigeration Systems

Refrigeration or Temperature Reduction Due to Expansion of the Process Fluid on Passing Through Valve/Choke or Turbine

□ Two Possible Causes

□ Joule-Thompson Effect

□ Without Work Done or Heat Transfer

□ Turbine Expansion

□ With Removal of Work from the Gas Stream

□ Types of Expansion Refrigeration Systems.

□ Valve or Choke Expander Cooling System

□ Turbine Expander Cooling System

□ Pressure Expander (Reducing) Cooling System

Ogbon Natural Gas Processing & Unit Slide #14

Typical LTS Schematic

Ogbon Natural Gas Processing & Unit Slide #15

Refrigeration Process Unit Operations

□ LPG Turbo-Expander Recovery Unit

□ Absorber process for C3+ recovery

Raw Pipeline

Natural Natural

Gas Gas

Expander C2

HEX

LPG

Deethanizer

HEX LPG Column

Absorber

C3 + C3+

C5 +

Ogbon Natural Gas Processing & Unit Slide #16

Refrigeration Process Unit Operations

□ Ethane Plus Expander Recovery Plant

□ Expander process for C2+ and NGL recovery.

Raw Pipeline

Natural Natural

Gas Gas

Ethane

Expander

Refrigerant Deethanizer

Deethanizer

NGL

Ogbon Natural Gas Processing & Unit Slide #17

Refrigeration Process Unit Operations

□ Nitrogen Rejection Unit

□ Nitrogen is one of the major contaminants which must be removed due to the fact that it lowers the calorific value of the natural gas and increases its transport volume

□ Nitrogen rejection units (NRUs) are used to remove inert components from the sales natural gas in natural gas processing plants to be able to meet customer and transmission pipeline standards

□ In the process, Nitrogen is normally vented into the atmosphere with small percentage of methane or re-injected for EOR projects.

□ There two types of NRU processes; the Solvent Absorption & Cryogenic Refrigeration processes

Ogbon Natural Gas Processing & Unit Slide #18

Refrigeration Process Unit Operations

□ Nitrogen Re-injection Unit

□ The Solvent Absorption Process

□ Methane and heavier hydrocarbons are separated from the nitrogen by a lean solvent

□ The methane and heavier hydrocarbons are flashed off from the rich solvent by depressurizing stream.

□ The lean solvent from the flash regeneration process is returned back to the top of the methane absorber.

Basic Process Flow Diagram of Natural Gas Absorption NRU

Lean

Nitroge

Solvent

n

Natural Flash Methan

Gas Rich

e&

Gas Solvent Regenerati

Ex-Well

Treatme on Heavie

nt s

Ogbon Natural Gas Processing & Unit Slide #19

Refrigeration Process Unit Operations

- Nitrogen Re-injection Unit
- Cryogenic Refrigeration NRU Process
- The cryogenic refrigeration Nitrogen Rejection Units(NRU) operate at cryogenic temperatures of -180 °C through the application of turbo-expanders.
- These condense out the methane and they are usually contained within a cold box.
- Other systems use other forms of heat exchangers for the same process
- There are several process configurations but the one NRU system illustrated here have its components are broadly divided into two sections; the prefractionation and the fractionation sections

Ogbon Natural Gas Processing & Unit Slide #20

Refrigeration Process Unit Operations

- Nitrogen Rejection Unit

Chart Energy & Chemical Inc. NRU Syste

Fractionation System

NRU Cold Box image

(Chart Energy & Chemical Inc.)

Ogbon Natural Gas Processing & Unit Slide #21

Refrigeration Process Unit Operations

- Nitrogen Re-injection Unit
 - Cryogenic Refrigeration NRU Process
 - The Prefractionator System
 - Typical feed to the prefractionator system is at 3000 kPag and it operates at about 2500 kPag and condenses out a portion of the incoming methane, thereby enriching the feed vapour in its nitrogen content up to 35%.
 - High pressure liquid methane residue is removed from the bottom and Vaporized to be sent to a methane residue compressor.
 - The NRU cold section is fed by the overhead from the prefractionator system
- Other advantages of the prefractionator are:
- A significant portion of the total methane residue product is produced at higher pressure, which results in lower methane

recompression energy.

□ Typically, greater than 50% of the total methane is recovered as its bottom product and this increases the capacity of the plant.

Ogbon Natural Gas Processing & Unit Slide #22

Refrigeration Process Unit Operations

□ Nitrogen Rejection Unit

Typical Pre-fractionating System of the Cryogenic NRU

Heat Fractionator

Exchanger System

Cold Section

NRU

Prefractionator

Pre-fractionating

System

Heat

Exchanger

Ogbon Natural Gas Processing & Unit Slide #23

Refrigeration Process Unit Operations

□ Nitrogen Re-injection Unit

□ Cryogenic Refrigeration NRU Process

□ The Fractionator System

□ The nitrogen-rich gas is fed into a heat exchanger, flashed and separated into two feeds before entering into the low pressure NRU fractionator

□ The fractionator column produces a high purity nitrogen distillate stream with a residual methane content ranging from 1 to 3%.

□ The nitrogen product is heated against warmer process streams before exiting the NRU.

□ The methane bottoms product typically contains between 2 to 5% nitrogen; however, a purity as low as 0.5% nitrogen is possible.

□ The methane residue liquid product serves as the major source of refrigeration for the NRU.

□ The bottoms product is usually pumped and then re-vaporized by cross exchange against the incoming feed stream.

□ The vaporized is compressed at the methane residue compressor.

Ogbon Natural Gas Processing & Unit Slide #24

Refrigeration Process Unit Operations

□ Nitrogen Rejection Unit

NRU Cold Box image

(Chart Energy & Chemical Inc.

Ogbon Natural Gas Processing & Unit Slide #25

Refrigeration Process Unit Operations

- Joule-Thomson Valve or Choke Refrigeration Unit Operations
- Temperature Change is Proportional to Pressure Drop.
- Process Fluid Vapour Condenses.
- Degree of temperature reduction and condensation proportional to operating condition and composition specification.
- Cooling Associated with Constant Enthalpy is Estimated from Correlation Chart

Ogbon Natural Gas Processing & Unit Slide #26

Refrigeration Process Unit Operations

- Joule-Thomson Valve or Choke Refrigeration Unit Operations
- Temperature Change is Proportional to Pressure Drop.
- Process Fluid Vapour Condenses.
- Degree of temperature reduction and condensation proportional to operating condition and composition specification.
- Cooling Associated with Constant Enthalpy is Estimated from Correlation Chart

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Ogbon Natural Gas Processing & Unit Slide #27

Refrigeration Process Unit Operations

- LNG Production Processing Plant
- Basic refrigeration systems employed in LNG production is the Expansion Refrigeration.
- The above is invariably combined with other methods of refrigeration such as mechanical refrigeration, J-T valve expansion, Turbo-Expansion, etc.
- The Following process flow diagrams illustrate the magnitude of the number of processes in the industry for LNG production.
- Basically the processes involve
- Pressure control – to bring stream process to plant pressure
- Condensate Removal
- Treatment processes
- Dehydration
- Precooling refrigeration
- Cryogenic Refrigeration
- NGL fractionation/LPG production

Ogbon Natural Gas Processing & Unit Slide #28

Refrigeration Process Unit Operations

- LNG Production Processing Plant

- Basic Processes of LNG Production: From Well Head to Shipment

Ogbon Natural Gas Processing & Unit Slide #29

Refrigeration Process Unit Operations

- The Bonny NLNG

- Precooling is by propane refrigeration

- Cryogenic Refrigeration(Liquefaction) by MR at the Main Cryogenic Heat Exchanger with Expander

- LPG injected to increase heating value

- LPG production through NGL fractionation

Gas Treatment & Precooling

Gas Treatment & Precooling

Ogbon Natural Gas Processing & Unit Slide #30

Absorption Refrigeration Unit

Operations

- LNG Regasification Process

- LNG needs to be regasified to natural gas at the customer's facility before compression into sales pipeline

- Major equipment of a regasification plant include

- Expander

- Demethanizer column

- Deethanizer column

- Compressor

Ogbon Natural Gas Processing & Unit Slide #31

Lesson IV - I

Natural Gas Processing and Unit Operations

(Chemical-GTL Processes)

Ogbon Natural Gas Processing & Unit Slide #32

Sub-Topic

Gas-To-Liquid (GTL) Processes

Ogbon Natural Gas Processing & Unit Slide #33

GTL Processing Plant

- Gas-To-Liquid(GTL) Technology

- This is the technology of producing liquid hydrocarbon fuels from two main sources;

□ Coal □ Coal-to-Liquid (CTL)

□ Natural gas □ Gas-To-Liquid (GTL).

□ Coal is oxidized under high pressure and temperature to produce synthetic gas (syngas) while the Natural gas is partially oxidized with air to produce synthesis gas (syngas).

□ GTL is therefore a generic name for production of hydrocarbon liquids from any hydrocarbon feed stock

□ The process using natural gas as the feedstock involves the conversion of the syngas into liquid hydrocarbons of various carbon-chain lengths through Fischer-Tropsch (FT) process

□ Fischer-Tropsch (FT) process rearranges carbon and hydrogen molecules in a manner that produces a liquid, heavier hydrocarbon molecules

Ogbon Natural Gas Processing & Unit Slide #34

GTL Processing Plant

□ Gas-To-Liquid(GTL) Technology

□ Lastly the heavy fraction of these product is separated and cracked back to transportation fuels in a hydrocracking reactor.

□ The liquid synthetic fuels (synfuels) produced are liquid hydrocarbons including paraffinic distillates and middle distillates consisting of naphtha, kerosene, and diesel that are then upgraded to meet marketability

□ Syngas can also go through the process that gives rise to oxygenate family such as methanol, dimethyl ether (DME), etc. and solid products manufactured from gas via gas conversion.

□ GTL technologies and Industry therefore embrace processes involved in the production of Hydrocarbon liquids and Oxygenates from Synthetic gas thereby giving rise to GTL-FT and GTL-Oxygenate.

Ogbon Natural Gas Processing & Unit Slide #35

GTL Processing Plant

Ogbon Natural Gas Processing & Unit Slide #36

GTL Processing Plant

□ GTL- Fischer-Tropsch (FT) Process

□ Syngas Production Reaction

Steam Reforming: $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$

FT Synthesis : $\text{CO} + 3\text{H}_2 \rightarrow \text{-(CH}_2\text{)} + \text{H}_2\text{O}$

Partial Oxidation: $\text{CH}_4 + 3/2\text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$

Water Gas Shift: $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$

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GTL Processing Plant

GTL Products.

Ogbon Natural Gas Processing & Unit Slide #38

GTL Processing Plant

GTL Products.

GTL-FT Plant Products compared with Refinery Products

Ogbon Natural Gas Processing & Unit Slide #39

GTL Processing Plant

□ Oxygenates Production Reaction

Methanol Production: $\text{CH}_4 + 1/2\text{O}_2 \rightarrow \text{CH}_3\text{OH}$

Ogbon Natural Gas Processing & Unit Slide #40

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Course:

PNG 516

Natural Gas Engineering II

Ogbon Natural Gas Engineering II Slide #1

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Ogbon Natural Gas Engineering II Slide #2

Lesson III

Application of Phase Behaviour in Natural Gas

Conditioning and Processing

Ogbon Natural Gas Engineering II Slide #3

Sub-Topics

□ Natural Gas Phase Behaviour

□ Equilibrium Calculation

□ Applications of Natural Gas Phase Behaviour on its Processing

Ogbon Natural Gas Engineering II Slide #4

Natural Gas Phase Behaviour

□ What is a Phase

□ A phase is homogenous and Physically
Distinct Part of a System with a 3 Phases of Water
Definite Bounding Surfaces (Interface).

□ Part of System Which Differs in its
Intensive Properties and Physically
Separable from Other Phases.

□ Bounding Surfaces(Interfaces) are
Determined by Condition of:

□ Pressure

□ Temperature

□ Different States of a Phase

□ Gaseous □ Water Vapour

□ Liquid □ Water

□ Solid □ Ice

□ □ Different Crystalline

□ Nature of Solid

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Natural Gas Phase Behaviour

□ What is a Phase

□ Phase Transition

Ogbon Natural Gas Engineering II Slide #6

Natural Gas Phase Behaviour

□ Phase Behaviour

□ The processing of natural gas invariably involves the
thermodynamic manipulation of its volume, temperature and
pressure conditions to “force” the gas into different phases or
quasi-phases which get separated in order to produce
merchantable hydrocarbons gas mixture containing mostly
methane.

□ It also involves the removal of contaminants.

□ The response of the natural gas upon subjection to changes in
temperature-pressure conditions and the quantities of the
phases(liquid & gas) produced with their respective compounds,
all depend on the phase behaviour of the natural gas.

□ The phase behaviour can therefore be defined as the changes in
phase exhibited by the natural gas when subjected to different
temperature-pressure conditions.

□ During production, the hydrocarbon molecule undergo various
phase- and property-altering intermediate stages which are

crucial in designing and operating the processes efficiently and optimally.

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Natural Gas Phase Behaviour

□ Phase Behaviour

□ Phase Behavior is the part of thermodynamics that gives us the tools for the complete understanding of how fluids behave at any of the stages of processing.

□ The study of natural gas behaviour is apt, essential and extremely important because the range of processing temperature and pressure is quite large (LNG -160 °C to GTL + 300 °C and 14.7 psia to 1200 psia) and within these ranges, petroleum can transcend the three phases of matter.

□ Phase behaviour is the exhibition of different phases of natural gas when subjected to changes in temperature and pressure.

□ Phase behaviour of natural gas is normally exhibited by its phase diagram and mostly understood through the study of the phase diagram.

□ The Phase Diagram is conditions of pressure and temperature under which different phases of substance can exist

Ogbon Natural Gas Engineering II Slide #8

Natural Gas Phase Behaviour

□ Overview of Phase Diagram

□ Temperature-Pressure Diagram illustrating various conditions (T-P) under which different phases of a system (substance) exist

□ Diagram that indicates whether substance exists as solid, liquid or vapour at any given temperature and pressure

□ A pictorial representation of the conditions of temperature and pressure for each phase of a pure substance.

□ The phase diagram also describes the temperatures and pressure where the phases are in equilibrium.

Pure Substance

(Water) Multi-components

Phase Diagram Phase Diagram

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Equilibrium Calculation

□ Phases in Equilibrium

□ Region between bubble point and dew point lines of the

phase diagram of a mixture

- Region where gas and liquid phases in equilibrium
- Quantities and compositions of phases are different for each set of pressure and temperature
- Features of System in Equilibrium
- System is at rest
- No changes taking place
- Inter-molecular forces of repulsion and attraction are equal
- Exists without change in:
 - Pressure
 - Temperature
 - Molecular attraction
 - Molecular repulsion

Ogbon Natural Gas Engineering II Slide #10

Equilibrium Calculation

- Equilibrium Calculation
- Calculations used to determine quantities and components of gas and liquid of mixture in equilibrium
- Uses of equilibrium calculation
- Prediction of behaviour and composition of reservoir fluid
- Determination of optimum conditions for processing reservoir fluid
- Equilibrium calculation methods
- Bubble point method
- Dew point method
- Within equilibrium region method
- Calculation of quantities and compositions of gas and liquid phases in equilibrium within the two phase region

Ogbon Natural Gas Engineering II Slide #11

Equilibrium Calculation

- Ideal Solution
- Properties of Ideal Solution
- Mutual solubility when mixed
- No chemical reaction within the mixture
- Equal molecular diameters
- Mix is uniform
- Inter-molecular forces of attraction and repulsion between liked and un-liked molecules are equal

- No heat generated or taken in the mixture
- Solution volume is sum of component volumes
- Mixture components behave same chemically and physically
- Examples of Ideal Solutions
- Gas mixtures at low pressures
- Liquid mixtures of same homogeneous series components (at low pressures)

Ogbon Natural Gas Engineering II Slide #12

Equilibrium Calculation

- Flash Evaporation Equilibrium Calculation
- Consider a liquid hydrocarbon system that has pressure reduction resulting in a flash vaporization resulting in vapour and liquid at equilibrium condition of P and T

Flash n_g ; y_j

Evaporation

n ; Z_j

n_L ; x_j

- Now consider the followings

- P & T

P_j = Partial Pressure of Component j in the Gas that is in Equilibrium with

- the Liquid in the ideal Solution Equilibrium

- X_j = Mole Fraction of Component j in Liquid (normally Given)

- y_j = Mole Fraction of Component j in Gas (normally Given)

- P_{vj} = Vapour Pressure the Pure Component j Would Exert at Temperature of interest

- n = Total Mixture Number of Moles

- n_L = Number of moles in the Liquid Phase

- n_g = Number of Moles in the Gas Phase

- z_j = Mole Fraction of j in Mixture

- $z_j n$ = Moles of j in Mixture n_g n_L

- $x_j n_L$ = Moles of j in Liquid Phase n_g n_L

- $y_j n_g$ = Moles of j in Gas Phase n n

Ogbon Natural Gas Engineering II Slide #13

Equilibrium Calculation

- Flash Evaporation Equilibrium Calculation

- For an Idea Mixture

The Equilibrium equation is as follows

Z_j Z_j

□ j □

X □

$\sum P_j$

and $\sum Y_j$

$\sum P$

$1 = n_g + 1 + n_L + 1$

$\sum P = \sum P$

where $\sum V_j$

$Y_j = P_j$

$X_j = P$

Above equation is in considering Just ONE Mole of Mixture

n_L = Fraction of ONE Mole of Mixture that is Liquid

L

n_L

n

n_g = Fraction of ONE Mole of Mixture that is Gas

n_g

n

$\sum$ For Non-Ideal Mixture

The Equilibrium equation is as follows

Z_j and

$\sum X_j = 1 = n_g + \sum K_j + 1$

$\sum Y_j = \sum$

Z_j

$\sum 1 = \sum$

$1 = \sum 1 + n_L$

where $\sum K$

$\sum j$

Y_j

$K_j =$

X_j

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Equilibrium Calculation

$\sum$ Use of the Flash Evaporation Equilibrium Calculation

P_{vj} = Vapour pressure pure component j is gotten from cox vapour-pressure chart

K_j = equilibrium ratio also known as k-value, equilibrium constant, etc. Is gotten from the k-value charts

$\sum$ Use is through trial & error solution by assuming either or until summation is 1 or 0.9999 (see example)

$n_g = n_L$

$\sum$ Used to find composition of gas and liquid phase in equilibrium

□ Used to calculate bubble point pressure of ideal liquid solution

□ Used to calculate dew point of ideal gas solution

Ogbon Natural Gas Engineering II Slide #15

Equilibrium Calculation

□ Basic Steps for X_j and Y_j Calculation and Example

□ With Given Equilibrium P and T Get Components P_{vj} from Vapour Pressure Curves

□ Assume Between 0 - 0.99

□ Solve for ϕ_{yi} Until

$L \text{ Summation} = 1$

□ Solve for =

□ For ideal solutions solve for

$n_g \text{ } 1 \text{ } n_L$

OR □ For ALL Components until

OR

$Z_j \text{ } Z_j$

$\phi_Y \text{ For } \phi_{Yj} \text{ } \phi \text{ } 1$

$j \text{ } \phi$

non-ideal

□ $P \text{ } \phi$ solutions solve

$Z \text{ For } \phi \text{ } P \text{ } \phi$

$1 \text{ } n_L \text{ } \phi$

j

$X \text{ } \phi \text{ } 1 \text{ } \phi$

$1 \text{ } n \text{ } \phi \text{ } 1 \text{ } \phi \text{ } P \text{ } j$

$\phi \text{ } P \text{ } \phi$

□ $P \text{ OR } 1 \text{ } n \text{ } \phi \text{ } \phi$ For ALL Components until

L

$\phi \text{ } 1 \text{ } \phi \text{ } V \text{ } j$

$\phi \text{ } V \text{ } j \text{ } \phi$

$\phi \text{ } V \text{ } j$

P

g

$\phi \text{ } \phi$

OR Z_j

$\phi X \text{ } j \text{ } \phi \text{ } \phi$

$\phi P \text{ } j \text{ } \phi$

$\phi 1$

□ Summary on n_g Choice 1 $\phi \text{ } n_g \text{ } \phi \text{ } V \text{ } 1 \text{ } \phi \text{ } \phi$

□ $P \text{ } \phi$

□ If Summation $\phi 1$ Choose Larger

Z_j Value Next

Z_j Z_j

X_j Y_j

If Summation $\neq 1$, Choose Smaller Y_j ≤ 1

$1 - n_g$

j

K_j $1 - 1$ Value Next 1

j $1 - n_g$ $1 - n_L$ $1 - n_L$ $1 - n_L$

K_j K_j

j

Z_j

X_j 1

$1 - n_g$ K_j 1

j

Ogbon Natural Gas Engineering II Slide #16

Equilibrium Calculation

Making Use of K-Values and Making Choice of n_g

Z_j

X_j $1 - n_g$ K_j 1

j

g_j

For X_j

If Summation $\neq 1$, Next n_g Larger

If Summation $\neq 1$, next n_g Smaller

The K-Value Should Be Experimentally Determined

Accurately for Real Behaviour of Solution to Be Reflected.

Quicken the Trial and Error Process, by getting a Value of n_L or n_g That Will Put Summation on Opposite Side of 1

When Summation Values on Both Sides Are Very Close to One, Apply Linear Interpolation/Extrapolation

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Equilibrium Calculation

Vapour Pressure - Temperature Plot for n-Paraffins

350

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Equilibrium Calculation

K-Value Chart for n-Butane

0.5

02

SLIDE 19

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Natural Slide #19

Equilibrium Calculation

Example 1

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In-Class Exercise

□ Solve the same problem in Example 1 assuming a non-ideal condition and using K-values

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Equilibrium Calculation

□ Bubble Point Pressure Calculation of Ideal Liquid Solution at a given Temperature

Composition Mixture @ PB

□ $n_L = n$ □ $n_g = 0$

□ Pressure of Mixture @ PB

□ $P = P_B$

□ Substituting in equilibrium calculation equation for ideal solution

 $\sum_j n_j Z_j$ □ □ □ $\sum_j Z_j P_j^v$ □ □ P_B □

0 □ □

□ P_j □□ V_j □□ n □ P □

□ □

□ P_j □□ V_j □ $= \sum_j P_j = 1$

b

□ □ □

 $P_B = \sum_j P_j V_j$ □ That is, for that temperature, P_B = Summation of Products of

□ Components Mole Fractions

and

□ Components Vapour Pressures

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Equilibrium Calculation

□ Dew Point Pressure Calculation of Ideal Liquid Solution for a Given Temperature

Composition Mixture @ PD

$$\square n_L = 0$$

$$\square n_g = n$$

□ Pressure of Mixture @ PD

$$\square P = P_D$$

□ Substituting in the equilibrium calculation equation,

$$Z_j n = Z_j P_D$$

$$\square \square Z_j P_D$$

$$\square \square P_j \square$$

$$0 \square \square \square V \square \square n$$

$$P$$

$$= \square \square V_j \square \square$$

$$\square \square$$

$$P$$

$$\square = 1$$

$$\square P \square \square P_D \square v_j$$

$$1$$

$$Z_j$$

$$\square P_D = \square P$$

$$V_j$$

□ P_D = Reciprocal of Summation of Quotients of Components

Mole Fraction Divided by Components Vapour Pressure

Ogbon Natural Gas Engineering II Slide #23

Equilibrium Calculation

□ Bubble Point of Real Solution

□ As Shown Earlier;

□ Composition Mixture @ PB

$$\square n_L = n$$

$$\square n_g = 0$$

□ Substituting in Equilibrium equation

$$\square \square y_j = \square Z_j K_j = 1$$

□ Impossible to Solve For Press Directly

□ Pressure is NOT Implicit in above equation

□ P_B Cannot be Solved for Directly

□ Pressure is Contained in K-Value

□ Trial and Error Process is Employed

Ogbon Natural Gas Engineering II Slide #24

Equilibrium Calculation

□ Making Use of K-Values and Making Choice of P_B for Bubble

Point Pressure Calculation

□ Trial and Error Process for PB

□ Select Trial Values of PB

□ Get Their Respective K-Values From Charts

□ Solve the bubble point equation Until Summation is 1

□ Guidelines

□ If $\sum Z_j K_j < 1$, Repeat with Lower Value of Trial PB

□ If $\sum Z_j K_j > 1$, Repeat with Greater Value of PB

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Equilibrium Calculation

□ Bubble Point of Real Solution

Example 2.

Estimate the bubble point pressure for the mixture in Example 1 at 150 °F.

□ Start with initial PB assuming liquid is ideal.

□ From Example 2-2, PB = 247 psia.

□ Get K_j Values from the chart appendix A

□ For P_b = 247 psia and T = 150 °F.

□ Calculate $\sum Z_j K_j$ Assume another PB

□ If $\sum Z_j K_j < 1$ Assume another PB and so on

Ogbon Natural Gas Engineering II Slide #26

Equilibrium Calculation

□ Bubble Point of Real Solution

Assumption 1: PB = 247 psia .

Composition K-Value at

Component of Mixture Mole at 247 psia & 150 °F

Fraction (Z_j) (K_j)

Z_j K_j

1. C₃ 0.61 1.270 0.775

2. n-C₄ 0.28 0.500 0.140

3. n-C₅C 0.11 0.203 0.022

$\sum Z_j K_j = 0.937$

$\sum Z_j K_j < 1$, Get PB₂ > PB₁

Assumption 2: PB = 200 psia

Composition K-Value at

Component of Mixture Mole 200 psia & 150 OF

Fraction (Zj) (Kj) Zj Kj

1. C3 0.61 1.520 0.927
2. n-C4 0.28 0.502 0.167
3. n-C5C 0.11 0.236 0.026

$$\sum Z_j K_j = 1.120$$

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Equilibrium Calculation

□ Bubble Point of Real Solution

□ Now Plot $\sum Z_j K_j$ Against P and Interpolate for P at $\sum Z_j K_j = 1$. See fig below

□ .

□ From Fig. 2-29, $\sum Z_j K_j = 1$ at 231 psia

□ Final Composition is As Follows

□ Composition K-value at

□ Component of Mixture Mole 231 psia & 150 OF

□ Fraction (Zj) (Kj)

Zj Kj

- 1. C3 0.61 1.350 0.824
- 2. n-C4 0.28 0.534 0.150
- 3. n-C5 0.11 0.223 0.025
- $\sum Z_j K_j = 0.999$

□ Note That $\sum Z_j K_j = \sum Y_j$;

□ This is Sum of the Components in the Small Gas Bubble at the bubble point.

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Equilibrium Calculation

□ Dew Point Pressure Calculation for Real Gas

□ Basic Assumption

□ $n_L = 0$ and $n_g = n$ □ .

□ Equilibrium Equation Reduces to

$$\sum X_j = \sum Z_j = 1$$

□ Kj

□ Note that PD is not apparent but imbedded in Kj, therefore several PDs have to be assumed and the equation solved for all components until an assumed PD gives

$$\sum X_j = 1$$

□ Example 3

□ Using Example 1 Values, Assume a Real Mixture and Calculate the Dew-point Pressure of the Mixture. Start With PD = 136 Psia As Already Calculated.

□ Composition K-value at

□ Component of Mixture Mole 142 Psia & 150 OF

□ Fraction (Zj) (Kj) Zj /

Kj

- 1. C3 0.61 2.100 0.290
- 2. n-C4 0.28 0.782 0.350
- 3. n-C5 0.11 0.312 0.352
- $\sum X_j = \sum Z_j/k_j = 1.000$

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Take Home Exercise

- Determine the bubble point and dew point pressures of the mixture in Example 2 at 100 OF

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Equilibrium Calculation

- Equilibrium calculation and the Phase Diagram
- . □ The nL and ng calculated for a particular T and P represent a point in the P-T phase diagram
- All P-T point at the same vapour-liquid proportions constitute a line in the phase diagram
- All such points for the bubble point and dew point constitute the bubble point and dew point lines respectively

Ogbon Natural Gas Engineering II Slide #31

Applications of Phase Behaviour in Natural Gas

Processing

- One of the first issues to be resolved by a facilities engineer working in a gas plant, gas production facility or transmission pipelines is where the process is operating with respect to the phase diagram of the natural gas of interest.
- This will enable the engineer to know if there will be liquid condensation or not; and also to determine the quantity of hydrocarbon and/or water liquid that will form
- The above will assist the engineer to determine
- If further optimization is needed and the level of optimization needed.
- The magnitude of the adverse effect of the liquid production to the process

Ogbon Natural Gas Engineering II Slide #32

Applications of Phase Behaviour in Natural Gas Processing

- Applications of Phase Behaviour in Natural Gas

Processing

Phases, Composition and conditions of the feed gas and

treated(lean) gas

Ogbon Natural Gas Engineering II Slide #33

Applications of Phase Behaviour in Natural Gas Processing

□ Application of Phase Behaviour in Pipeline Gas

□ The phase diagram provides a

Transmission

means for predicting whether multi-

phase flow conditions are to be

found

□ An upper and lower set of operational conditions allowable within the pipeline (in terms of pressure and temperature) will exist.

□ These will dictate the maximum pressure that the pipe can endure without failure (i.e., maximum operating pressure) and lowest pressure conditions that will be

□ If the operating assigned envelope agreement by contractual is as indicated in the above figure it means with the that the gas fig) end consumer(see must have the phase diagram of Gas A to be dry.

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□ Every other phase envelope

Natural will produce

Gas Engineering II liquid Slide #34

Applications of Phase Behaviour in Natural Gas Processing

□ Applications of Phase Behaviour in Separation Process

□ Generally, separation process of the reservoir fluid takes two or three stages of separation depending on the initial inlet conditions of T & P and the required products

□ The phase behaviour equilibrium equation is employed to determine mole fractions of vapour() and liquid() flashed out of each stage of separation which are used to determine;

□ Optimum separationng conditions:nL separator pressure and temperature

□ Compositions of the separated gas and oil phases

- Oil formation volume factor
- Producing Gas-Oil ratio
- API gravity of the stock tank oil

Ogbon Natural Gas Engineering II Slide #35

Applications of Phase Behaviour in Natural Gas Processing

- Applications of Phase Behaviour in Separation Process
- Oil formation volume factor
- The oil formation volume factor(B_o) calculation involves the following steps:

1. Calculate the Molecular weight of the STO

$$M_{STO} = \sum x_j M_j$$

2. Calculate the weight of C_3+ (M_{C_3+}) and use this with the respective component densities to calculate liquid volume of C_3+ $\sum x_j M_j$

$$V_{LC_3} = \frac{M_{C_3+}}{\rho_{C_3+}}$$

$$\rho_{C_3+}$$

3. Calculate the density of C_M

$$\rho_{C_3+} = \frac{M_{C_3+}}{V_{LC_3}}$$

$$\rho_{C_3+}$$

$$V_{C_3+}$$

$$\rho_{STO}$$

4. Make corrections for C_2 and C_1 (see McCain) to get

5. Calculate STO Density ρ_{STO}

$$\rho_{STO} = \frac{M_{STO}}{V_{STO}}$$

$$62.4$$

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Applications of Phase Behaviour in Natural Gas Processing

- Applications of Phase Behaviour in Separation Process
- Oil formation volume factor
- For the Two-Stage Separation

$$MOR = \frac{V_{STO}}{V_{G_1}}$$

$$BO = \frac{res. \text{ bbl}}{STB}$$

$$\rho_{OR} = \frac{M_{STO}}{V_{L1} + V_{L2}}$$

$$\rho_{STO} = \frac{M_{STO}}{V_{STO}}$$

$$0.1415$$

$$API = \frac{141.5}{\rho_{STO}}$$

$$\rho_{STO}$$

$$\rho_{STO} = \frac{M_{STO}}{V_{STO}}$$

$$\rho_{STO} = \frac{M_{STO}}{V_{STO}}$$

$$\rho_{STO} = \frac{M_{STO}}{V_{STO}}$$

$$\rho_{STO} = \frac{M_{STO}}{V_{STO}}$$

2,138.ng1 .□ STO

RSP1 □

nL1 .nL2 . M STO

Stock Tank Separation (RST)

2,138.ng2 .□ STO

RST □

nL2 .MSTO

Total GOR(RT)

RT □RSP1□ RST

Ogbon Natural Gas Engineering II Slide #37

Applications of Phase Behaviour in Natural Gas Processing

□ Applications of Phase Behaviour in Separation Process

□ Optimum Separator Operating Condition

□ The Flash Evaporation Equilibrium calculation is carried out for several T & P and the API, Vo, GOR and Bo are determined and plotted

□ This is now done with computer applications such as HYSIS, Flash< et

□ At Optimum Conditions

□ Oil API Value Is Maximum

□ Stock Tank Oil Quantity is Maximum/Optimum

□ GOR is Minimal

□ Formation Volume Factor is Minimal

Ogbon Natural Gas Engineering II Slide #38

Applications of Phase Behaviour in Natural Gas Processing

□ Applications of Phase Behaviour in Determining Effect of Hydrate and Water in Natural Gas Processing

Water = 0.06% Water = 0.0427%

Water = 0.06% Water = 0.0427- 0.00422%

Ogbon Water = 0.0427% Natural Gas Engineering II Slide #39

Applications of Phase Behaviour in Natural Gas Processing

□ Applications of Phase Behaviour in Determining Effect of Hydrate and Water in Natural Gas Processing

Ogbon Natural Gas Engineering II Slide #40

Applications of Phase Behaviour in Natural Gas Processing

□ Applications of Phase Behaviour in Determining Effect of Hydrate and Water in Natural Gas Processing

□ Phase Behaviour is usually applied to determine the occurrence of hydrates

Water/Hydrocarbon (HC) System Phase Diagram

Q2

Q1

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Applications of Phase Behaviour in Natural Gas

Processing

□ Applications of Phase Behaviour in Determining Effect of Hydrate and Water in Natural Gas Processing

□ Major Features of the Water/Hydrocarbon (HC) System Phase Diagram

□ Critical point is point C □ condition where the liquid and gas hydrocarbon merge into a single hydrocarbon phase in equilibrium with liquid water.

□ Point Q2 □ upper quadruple point, where four phases (liquid water, liquid hydrocarbon, gaseous hydrocarbon, and solid hydrate) are in equilibrium.

□ Point Q1 □ lower quadruple point, typically occurs at 32 °F (ice freezing point) where four phases (ice, hydrate, liquid water, and hydrocarbon gas) are in equilibrium.

Line Q1 – Q2 □ represents the conditions for hydrate formation or dissociation

□ At these point, phases are not pure as Phase Behavior thermodynamics methods such as the Katz, Solid K-value equilibrium they contain some amount of the calculation, etc are normally applied to determine line Q1 – Q2 other substances according to their mutual solubility

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Applications of Phase Behaviour in Natural Gas

Processing

□ Applications of Phase Behaviour in Refrigeration

□ Vapour Compression Refrigeration

□ J-T Valve Expansion Refrigeration

□ Turbo-Expansion Refrigeration

□ LNG Production Process

□ The application of the phase diagram in the above refrigeration system is typified by the refrigerant cycle in the vapour-compression

refrigeration system shown below

2

1

3

5 4

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Course:

PNG 520

Natural Gas Processing and Unit

Operations

Ogbon Natural Gas Processing & Unit Slide #1

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Ogbon Natural Gas Processing & Unit Slide #2

Lesson IV - II

Natural Gas Processing and Unit Operations

(Adsorption Processes)

Ogbon Natural Gas Processing & Unit Slide #3

Adsorption Process Unit Operations

□ Natural Gas Adoption Processes

□ This is the removal of impurities from natural gas by solid(adsorbents) or desiccants

□ Adoption processes are employed to remove

□ Water □ Dehydration

□ Acid gas

□ Mercaptans

□ Helium

□ Mercury

□ Natural gas adoption processes are normally by batch processes

Sweetening

□ Batch Processes

□ Reactant is Discarded After Use.

□ Used for Limited to Small Amount of Sulfur Removal.

□ Applicable to Low Gas Flow Rates and Low H₂S Concentration.

□ Types of Batch Processes

- Iron Sponge
- Chemsweet.
- Sulfa-check
- Caustic Soda
- Molecular Sieves

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Adsorption Process Unit Operations

- Adsorption Dehydration Process
- This is the removal of water from natural gas by solid(adsorbents) or desiccants
- Basic desiccants used include:
 - Silica gel (most commonly used)
 - Molecular sieve Water Adsorption by
 - Activated alumina Desiccant Pellets
 - Activated carbon.
- Plant uses Batch Process

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Adsorption Process Unit Operations

- Adsorption Dehydration Process

Skid Mounted Dehydration Batch Process Unit Operation

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Adsorption Process Unit Operations

- Natural Gas Adsorption Sweetening Processes
- Sweetening processes remove acid gases to be able to reduce the natural gas toxicity and corrosion tendencies
- Generally, natural gas is expected to have impurities to these levels to be considered pure
- Max. Allowable Content(including N₂ as Inert Gases) = 2-3%
- N₂, H₂S and CO₂ are Also Removed For Their Economic Value
- Pipeline and Industry Specification
- 0.25g/100 ft³ Gas
- 4 ppm

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Adsorption Process Unit Operations

- Sweetening by Molecular Sieves
- Molecular sieves Sweeten and Dehydrate By Adsorption

- Molecular sieves are Non-toxic and Non-corrosive Alkali Metal Aluminosilicates that contain Intercrystalline voids which contaminant Molecules are adsorbed Into
- Inter-crystalline Voids about 11.40Å But Effective Void and Max. Adsorbable Molecule Diameter Depends on Base Structure Cation. (see Table 4-5)
- Adsorption of Aromatics, H₂S and Mercaptans From Liquids and Gas
- X-type Molecular Sieve Have More Adsorption Capacity.
- Molecular Sieves Adsorbs Only Polar Molecules and Capacity Increases with Degree of Polarity and H₂O Is Mostly Adsorbed Followed By Mercaptans, H₂S and CO₂
- Sulfur Contaminants Selective Removal is Achieved Due Degree of Affinity for Adsorption.

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Adsorption

Process Unit Operations

Molecular Sieves Types, Base Cation & Contaminants

Absorbed

□ .

Table 4-5A

Molecular Base Cation Adsorbable Molecule

Contaminants

Sieve Type Diameter- Å

Removed

3A Potassium 3

H₂O and NH₄

(Basically for

Dehydration)

4A Sodium 4 H₂S, CO₂,

SO₂, C₂H₄,

C₂H₆ &

C₃H₆

5A Calcium 5

Separates N-paraffins From

Iso- and

Cyclic Paraffins

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Adsorption Process Unit Operations

□ .

Ogbon Natural Gas Processing & Unit Slide #10

Adsorption Process Unit Operations

Typical Solid Desiccant Dehydration Plant

- Components
- Contractor (Absorber or Sorber)
- Has beds of Granular Desiccants Where Adsorption Occurs.
- Fluid Inlet and Outlet Connections.
- Flow is Down the Column; Reduces Disturbances.
- Filter Separator
- Removes all Solids and Contaminants.
- Regeneration Gas Heater.
- Produces Hot Regeneration Gas
- Regeneration Gas Cooler
- Cools the Rich Regeneration Gas.
- Regeneration Gas Scrubber.
- Removes the Water from the Regeneration Gas.
- Produces the Cool Gas for Contractor Beds Cooling.

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Adsorption Process Unit Operations

Typical Solid Desiccant Dehydration Plant Process Flow Diagram

Ogbon Natural Gas Processing & Unit Slide #12

Adsorption Process Unit Operations

Typical Solid Desiccant Dehydration

Inlet Gas Dehydration

Plant Cycle

- Rich Inlet Gas Stream Goes □ Inlet gas Flows Contactor
- Through Filter For Contaminant From Top, Goes Through
- Removal. Desiccant Beds and Got its
- Water Removed.

□ .

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Adsorption Process Unit Operations

Typical Solid Desiccant Dehydration

Desiccant Beds Regeneration

Plant Cycle

- Hot Regeneration Gas From □ Boiling and Evaporation Starts
- Heater is Released at the End at 240 °F and Continues Till
- of Dehydration Cycle to 350-375 °F for 4 Hours.
- Remove All Water From Beds
- Flowing from Bottom to Top. □ While One Adsorber is

Dehydrating, The Other is

□ .

Being Regenerated.

□ .

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Adsorption Process Unit Operations

Typical Solid Desiccant Dehydration

Desiccant Beds Cooling

Plant Cycle

□ Hot Regenerated Desiccant Beds □ The Cool and Hot Regeneration are Cooled by Shutting off or Gas Finally Goes Through the Bypassing the Heater. Regeneration Cooler and

□ Cool Regeneration Gas from Scrubber for All Adsorbed Water to Condense Out.

Scrubber then Flows From Top

□ Power Operated Valves, Downwards to Cool Beds.

Activated by a Timing Device, Cooling Terminate at 125 0F.

□ . Switch the Adsorber Between Dehydration, Regeneration and Cooling Steps

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Adsorption Process Unit Operations

□ Solid Desiccant Dehydration Plant

□ Major Points of Consideration.

□ Efficiency Decreases With Each Regeneration.

□ Plant is Always Put in Operation More Quickly after Shut Down.

□ Plant can be Adopted For Hydrocarbon Liquid Recovery.

□ Removal of all Contaminants Must be Ensured.

□ Operating Life of Desiccants is Between 1- 4 Years.

□ Sudden Pressure Surges Should be Avoided.

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Adsorption Sweetening Process Unit

Operations

□ Adsorption Sweetening Processes Consist of Mainly Chemicals That Absorb

Acid Gases.

☐ Factors Guiding Choice of Reactants.

☐ Cost ☐ Should Be Inexpensive

☐ Non-Hazardous.

☐ High Capacity and Affinity for Acid Gases.

☐ Reactant Waste Should Be Environmentally Safe.

☐ Types of Batch Processes.

☐ Molecular Sieves Sweetening Plant

☐ Iron Sponge Sweetening Process.

☐ Liquid and Slurry Processes

☐ Iron Oxide Slurry

☐ Chemsweet

☐ Sulfa-Check

☐ Caustic Soda

☐ Mercaptans be Removed and Spent Product Sold to Paper Co.

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Adsorption Sweetening Process Unit

Operations

☐ Advantages of Batch Process

☐ Relative Low Capital Investment.

☐ .

☐ Adequate Removal of Organic Sulfur Contaminants Such as Mercaptans.

☐ Low to Medium H₂S Conc.

☐ Removal With no Consumption of Reactant by CO₂.

☐ Process Has High Affinity for Sulfur Containing Gases and is Largely Independent of Operating Pressure

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Adsorption Sweetening Process Unit

Operations

☐ Molecular Sieves Sweetening Plant

Sour

Gas Enters Tower From Top, Gets Its Contaminants Removed at the Molecular Sieves and Goes Out Through the Bottom.

☐ Tower Is Regenerated With Hot Regeneration Gas From Bottom and Acid Regenerated Gas Sent for Flaring or Processing.

Molecular Sieves Sweetening Plant

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Adsorption Sweetening Process Unit

Operations

□ Iron Sponge Sweetening Process

□ Units.

□ The main processing agents are

Wooden Chips or Shavings

Impregnated With Hydrated Ferric

Oxide(Fe_2O_3)

□ Sodium Carbonate Added to Control

pH.

□ Process

□ Process Gas Enters From Top and

Gets Distributed by an Inlet

Diverter and Flows Downwards onto

the Hydrated Ferric Oxide

Impregnated Chips.

□ Distributed Gas Passes Through and

Reacts with Sponge and Leaves

through The Bottom

Ogbon Natural Gas Processing & Unit Slide #20

Adsorption Sweetening Process Unit

Operations

□ Iron Sponge Sweetening Process Units.

□ Process Reaction

□ .

□ $\text{Fe}_2\text{O}_3 + 3\text{H}_2\text{S} \rightarrow \text{Fe}_2\text{S}_3 + 3\text{H}_2\text{O}$

□ (Ferric Sulfide)

□ $\text{Fe}_2\text{O}_3 + 6\text{RSH} \rightarrow 2\text{Fe}(\text{RS})_3 + 3\text{H}_2\text{O}$

□ (Mercaptans) (Ferric Mercatide)

□ Process Regeneration.

Air or Oxygen Injected Into Inlet Gas and Fe_2S_3 is Immediately

Oxidized to Fe_2O_3 as Soon as H_2S is Absorbed.

□ $2\text{Fe}_2\text{S}_3 + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 6\text{S}$

□ $2\text{Fe}_2(\text{RS})_3 + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 3\text{RS}_2\text{SR}$

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Adsorption Sweetening Process Unit

Operations

□ Liquid and Slurry Sweetening Process Units

□ .

Iron Oxide Slurry

☐ ChemsweetSulfa-Check

☐ Caustic Soda

☐ Uses Mixtures of

☐ Water

☐ Methanol

☐ Formaldehyde

☐ Examples

☐ Di-chem, Inhibit & Scavinox

☐ Reacts with H₂S to Form

☐ FORMTHIONALS

☐ Components

☐ Contact Tower With Outside Flow Connections.

☐ Gas Distributor.

☐ Mist extractor

☐ Mixture

☐ Slurry of Zinc Oxide, Zinc Acetate and a Dispersant

☐ ZnO + ZnAc₂ + Dispersant

Ogbon Natural Gas Processing & Unit Slide #22

Adsorption Sweetening Process Unit

☐ Process.

Operations

☐ Liquid and Slurry Sweetening Process Units

☐ .

☐ Sour Gas Entering From Bottom Gets Dispersed into Smaller Bubbles by the Distributor.

☐ Bubbles Rise Through the Down Flowing Slurry, Get Mixed Up with Slurry and Get Sweetened.

☐ Gas Flow through Mist Extractor and Out.

☐ Sweetening

☐ ZnAc₂ + H₂S → ZnS + 2HAc

☐ Regeneration

☐ ZnO + 2HAc → ZnAc₂ + H₂O

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Adsorption Sweetening Process Unit

☐ Comparison Operations

of Batch Processes

Ogbon Natural Gas Processing & Unit Slide #24

Adsorption Sweetening Process Unit

□ Sweetening Operations

Process Selection Chart

□ .

□ .

Ogbon Natural Gas Processing & Unit Slide #25

Sulfur Recovery Process Unit

□ Modified Claus Processes

Operations

□ Modified Claus Once-Through Process

□ This involves direct oxidation of H₂S to Sulphur

□ The natural gas is mixed with a stoichiometric quantity of air to convert the incoming H₂S to elemental sulphur via direct oxidation.

□ Sulphur recovery efficiency is further increased through the application of the sub-dewpoint principle in a catalyst reactor below the sulphur dew point..

□ Direct oxidation of H₂S

$2\text{H}_2\text{S} + \text{O}_2 = 2/x \text{ S}_x + 2\text{H}_2\text{O} + \text{heat of reaction}$

□ Claus reaction

$2\text{H}_2\text{S} + \text{SO}_2 = 3/x \text{ S}_x + \text{H}_2\text{O} + \text{heat of reaction}$

where x = 2, 4, 5, 6, 8 is sulphur molecules of different sizes, according to temperature.

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Sulfur Recovery Process Unit

Operations

□ Modified Claus Processes

□ Modified Claus Once-Through

Process

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